MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, D TYPE, DIGITAL, KEY CODE KBR

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic D type flip-flop module

* 1.2 Classification This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2 APPLICABLE DOCUMENTS

* 2 1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.46 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KBR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/32-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0 5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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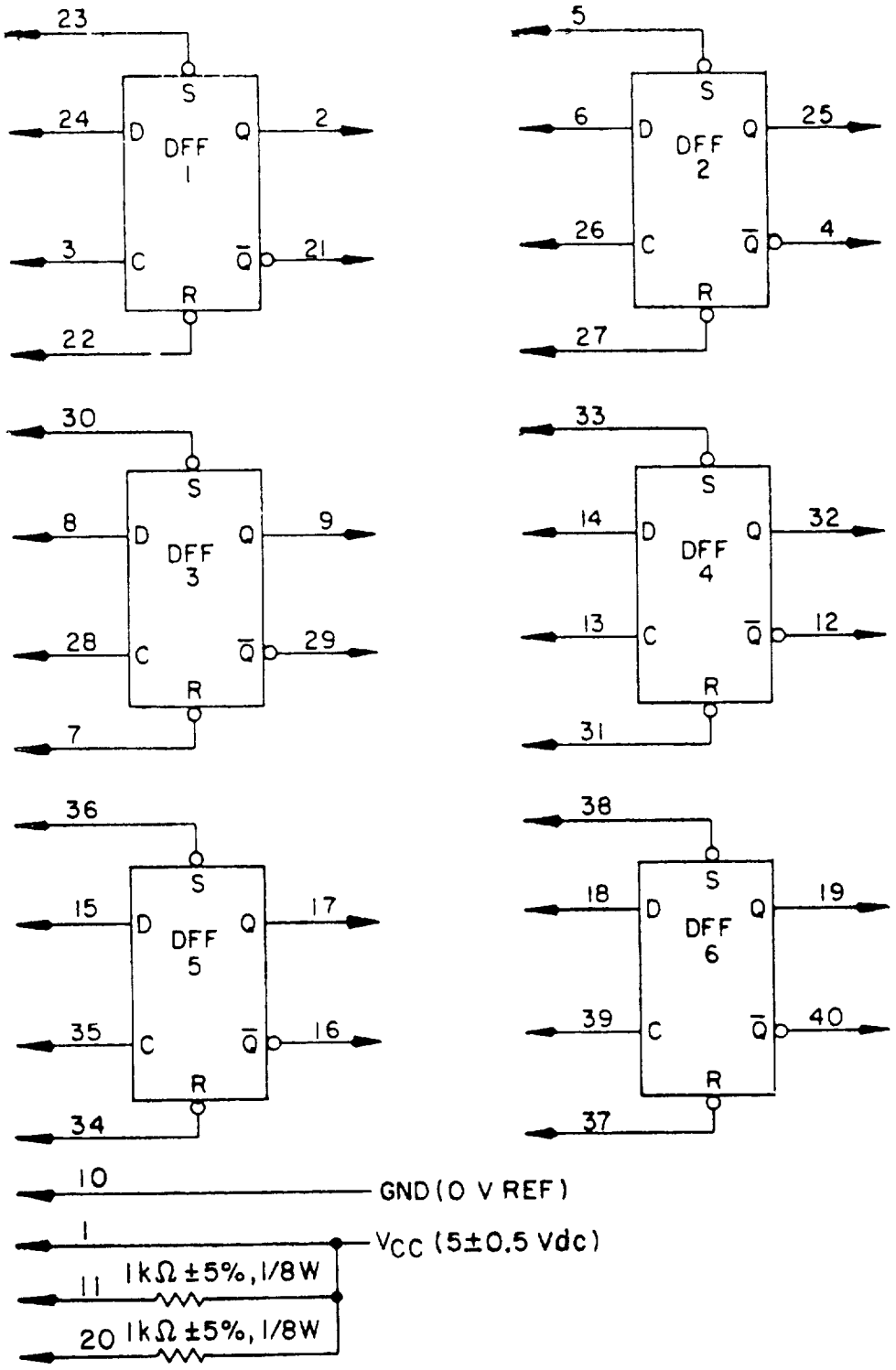


FIGURE 1 Module function diagram, D type flip-flop

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	III	S	0.01	0.6	0.01	0.6	0.01	0.6	0.01	0.6	V
B1-B6	Input breakdown current, data	IV	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	mA
B7-B18	Input breakdown current, clock and set	IV	S	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	mA
B19-B24	Input breakdown current, reset	IV	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	mA
C	Isolation	V	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
D1-D24	Input capacitance	VI	Q	2	15	2	15	2	15	2	15	pF
E1-E24	Static output voltage, low level	VII	X	0.0	0 3	0.0	0.3	0.0	0.3	0.0	0.3	V
F1-F36	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G6	Input current, data, low level	VIII	X	0.05	0.20	0.05	0.20	0.05	0.20	0.05	0.20	mA
G7-G18	Input current, clock and set, low level	VIII	X	0.05	0.40	0.05	0.40	0.05	0.40	0.05	0.40	mA
G19-G24	Input current, reset, low level	VIII	X	0.05	0.60	0.05	0.60	0.05	0.60	0.05	0.60	mA
H1-H6	Input current, data, high level	IX	X	0.0	10.0	0.0	10.0	0.0	10.0	0.0	10.0	µA
H7-H24	Input current, clock and set, high level	IX	X	0.0	20.0	0.0	20.0	0.0	20.0	0.0	20.0	µA
H25-H30	Input current, reset, high level	IX	X	0.0	30.0	0.0	30.0	0.0	30.0	0.0	30.0	µA
J1-J12	Output short circuit current	X	X	3.0	15.0	3.0	15.0	3.0	15.0	3.0	15.0	mA
J13,J14	Output short circuit current, resistors	X	X	5.1	5.9	5.1	5.9	5.1	5.9	5.1	5.9	mA
K1	Power supply current, high level	XI	S	1.0	9.0	1.0	9.0	1.0	9.0	1.0	9.0	mA
K2	Power supply current, low level	XI	S	1.0	9.0	1.0	9.0	1.0	9.0	1.0	9.0	mA
L1-L12	Transition time (t_{TLH}), C_L = 150 pF	XII	Q	10	100	10	100	10	100	10	100	ns
L13-L24	Transition time (t_{TLH}), C_L = 50 pF	XII	S	5	60	5	60	5	60	5	60	ns
M1-M12	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	5	60	5	60	5	60	5	60	ns
M13-M24	Transition time (t_{THL}), C_L = 50 pF	XIII	S	3	30	3	30	3	30	3	30	ns
N1-N12	Propagation delay time (t_{PHL}), clock-to-output, C_L = 150 pF	XIV	Q	25	200	25	200	25	220	25	220	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N13-N24	Propagation delay time (t_{PHL}), clock-to-output, C_L = 50 pF	XIV	S	25	150	25	150	25	170	25	170	ns
P1-P12	Propagation delay time (t_{PLH}), clock-to-output, C_L = 150 pF	XV	Q	25	140	25	140	25	150	25	150	ns
P13-P24	Propagation delay time (t_{PLH}), clock-to-output, C_L = 50 pF	XV	S	25	100	25	100	25	110	25	110	ns
R1-R12	Data setup time (maximum clock frequency, minimum clock pulse width) check	XVI	S	780	840	780	840	780	840	780	840	ns
S1-S12	Data hold time check	XVII	S	780	840	780	840	780	840	780	840	ns
T1-T24	Output crosstalk, low level and high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
V1-V12	Propagation delay time (t_{PHL}), set and reset to output, C_L = 150 pF	XX	Q	10	200	10	200	10	225	10	225	ns
V13-V24	Propagation delay time (t_{PHL}), set and reset to output, C_L = 50 pF	XX	S	10	150	10	150	10	175	10	175	ns
W1-W12	Propagation delay time (t_{PLH}), set and reset to output, C_L = 150 pF	XXI	Q	10	125	10	125	10	150	10	150	ns
W13-W24	Propagation delay time (t_{PLH}), set and reset to output, C_L = 50 pF	XXI	S	10	75	10	75	10	100	10	100	ns
X	Power supply filter capacitance	XXII	S	1.6	2.8	1.6	2.8	1 6	2.8	1.6	2.8	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through X shall be performed in the order specified in 4 4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification Part number M28787/32-1 supersedes part number M28787/32. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation (flip-flops only)	50 mW (outputs not loaded and V_{CC} = 5.5 V)
Supply voltage	5.0 Vdc ±10 percent
Clock input	
Frequency	2.5 MHz maximum (C_L = 50 pF)
Pulse width (high level and low level)	200 ns minimum
Triggering edge	Positive

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Data input current		
Low level		
(V_{IL} = 0.3 V)	-0.18 mA maximum	
(V_{IL} = 0 V)	-0.2 mA maximum	
High level		
(V_{IH} = 2.4 V)	10 µA maximum	
(V_{IH} = 5.5 V)	100 µA maximum	
Clock input current		
Low level		
(V_{IL} = 0.3 V)	-0.36 mA maximum	
(V_{IL} = 0 V)	-0.40 mA maximum	
High level		
(V_{IH} = 2.4 V)	20 µA maximum	
(V_{IH} = 5.5 V)	200 µA maximum	
Set input current		
Low level		
(V_{IL} = 0.3 V)	-0.18 mA maximum	
(V_{IL} = 0 V)	-0.4 mA maximum	
High level		
(V_{IH} = 2.4 V)	20 µA maximum	
(V_{IH} = 5.5 V)	200 µA maximum	
Reset input current		
Low level		
(V_{IL} = 0.3 V)	-0.36 mA maximum	
(V_{IL} = 0 V)	-0.60 mA maximum	
High level		
(V_{IH} = 2.4 V)	30 µA maximum	
(V_{IH} = 5.5 V)	300 µA maximum	
Input voltage		
Low level	-0.5 V minimum, 0.7 V maximum	
High level	2 V minimum, V_{CC} maximum	
Output voltage		
Low level (I_{OL} = 2.0 mA)	0.3 V maximum	
High level (I_{OH} = -100 µA)	2.4 V minimum	
Output drive maximum capability		
Low level (V_{OL} = 0.3 V)	2.0 mA	
High level (V_{OH} = 2.4 V)	-100 µA	
Propagation delay time	(C_L = 150 pF)	(C_L = 50 pF)
Clock to output		
t_{PLH}	150 ns maximum	110 ns maximum
t_{PHL}	220 ns maximum	170 ns maximum
Set and reset to output		
t_{PLH}	150 ns maximum	100 ns maximum
t_{PHL}	225 ns maximum	175 ns maximum
Set and reset pulse width (low level)	100 ns minimum	
Data input setup time	50 ns minimum	
Data input hold time	25 ns minimum	

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6.5 Module function. This module contains six independent low power D type flip-flops.

6.5.1 Input to output relationship. Each flip-flop has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 5/

t_n	t_{n+1}	
Input D	Output Q	Output $\overline{Q}$
L	L	H
H	H	L

1/ t_n = Time period before clock pulse.
2/ t_{n+1} = Time period after clock pulse.
3/ Low-level input to SET input sets Q output to a high level
4/ Low-level input to RESET input sets Q output to a low level.
5/ SET and RESET are independent of clock input

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Up-level source. Two independent 1 kΩ resistors are provided on the module for a convenient up-level source for unused inputs

6.5.4 Operation. This is an edge triggered D type flip-flop with independent set and reset inputs and complementary Q and $\overline{Q}$ outputs. Input information on the data input is tranferred to the output on the positive edge of the clock pulse. Clock triggering occurs at a voltage level during the rise transition of the clock pulse. After the clock input threshold voltage has been passed, the data input is locked out. The set and reset inputs are independent of the clock. A low-level input to the set line sets Q to a high level, and a low-level input to the reset line sets Q to a low level.

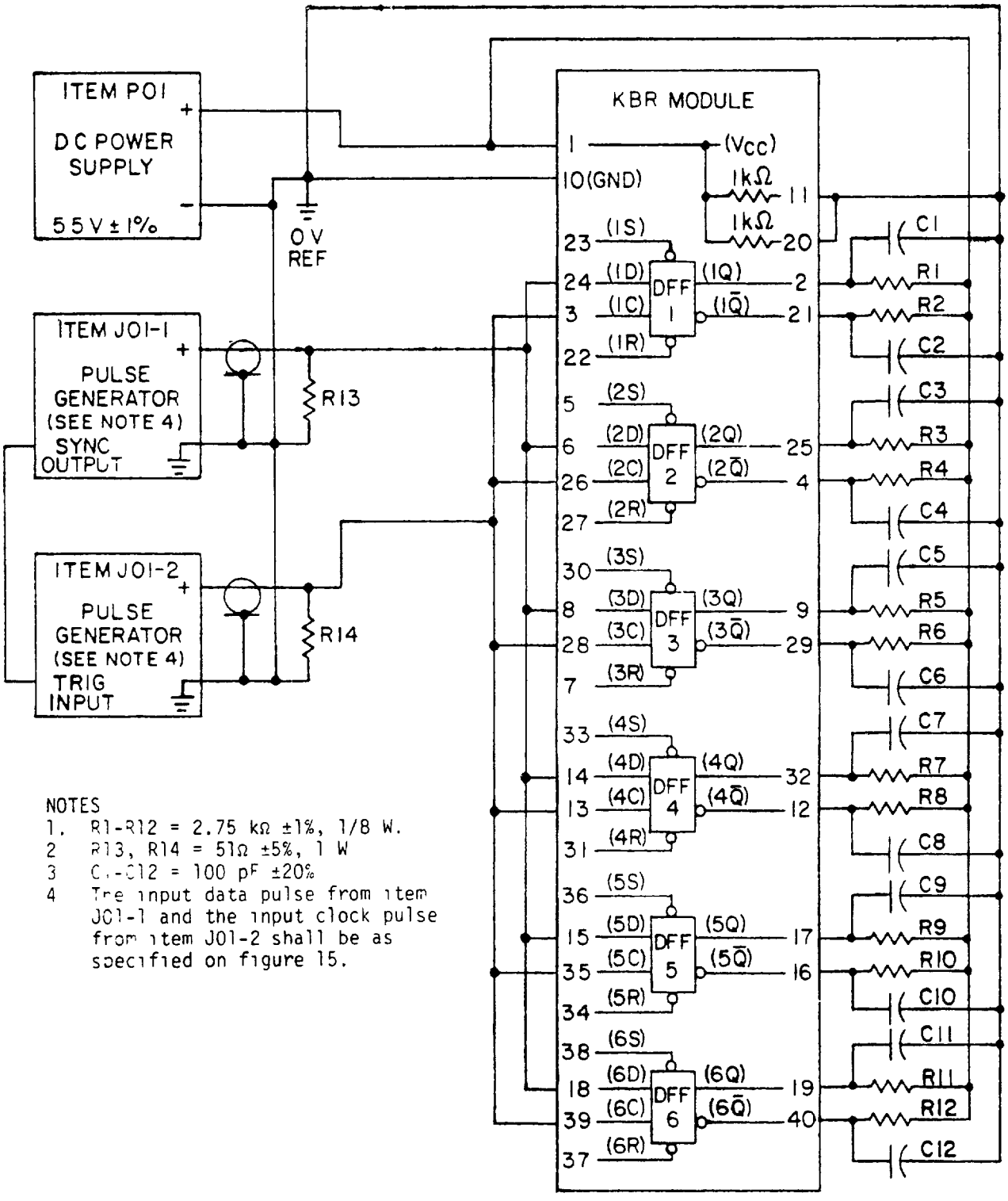
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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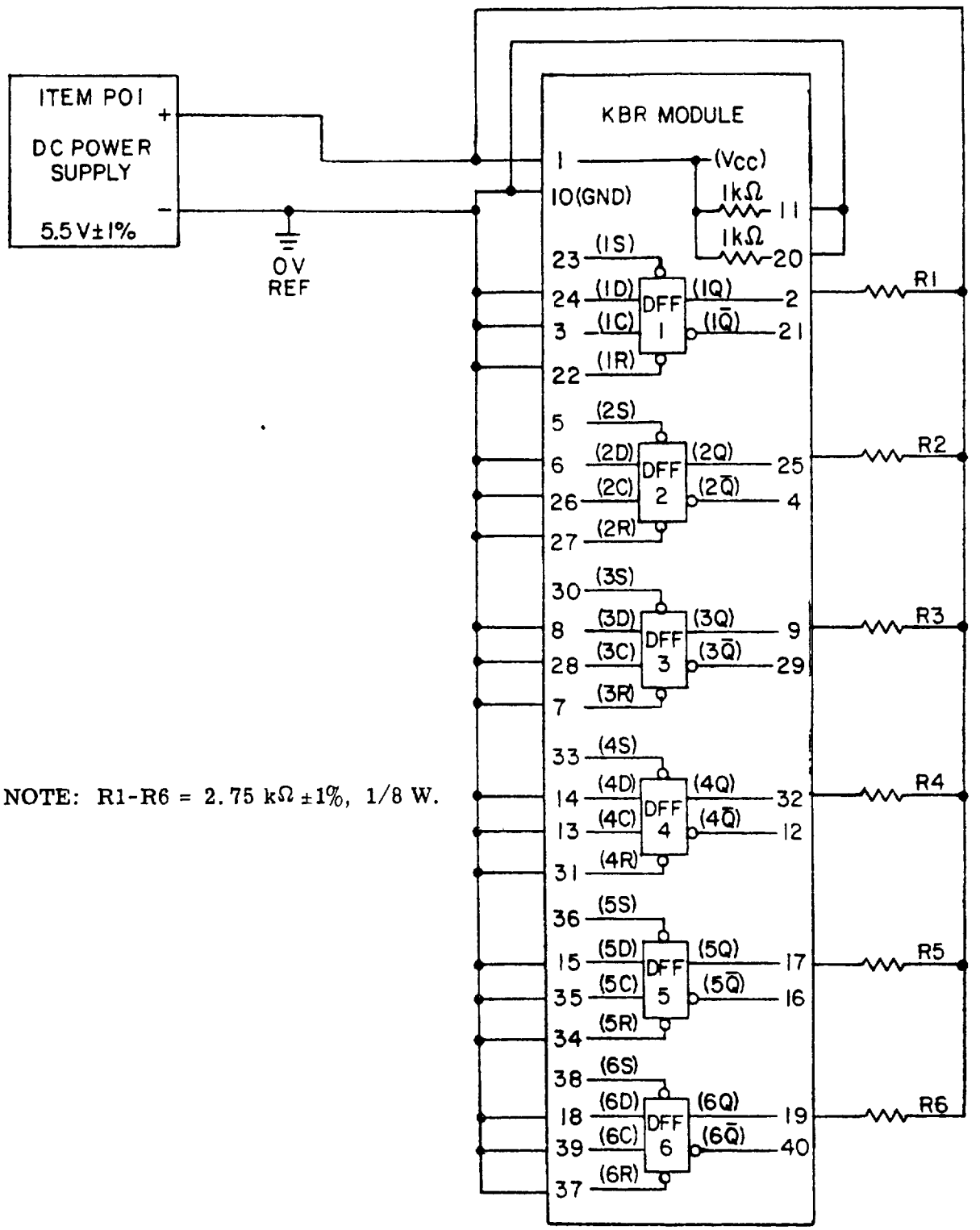
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NOTES
1. R1-R12 = 2.75 kΩ ±1%, 1/8 W.
2. R13, R14 = 51Ω ±5%, 1 W
3. C1-C12 = 100 pF ±20%
4. The input data pulse from item J01-1 and the input clock pulse from item J01-2 shall be as specified on figure 15.

FIGURE 2 Operating temperature and life test circuit.

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NOTE: R1-R6 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Transient thermal test circuit.

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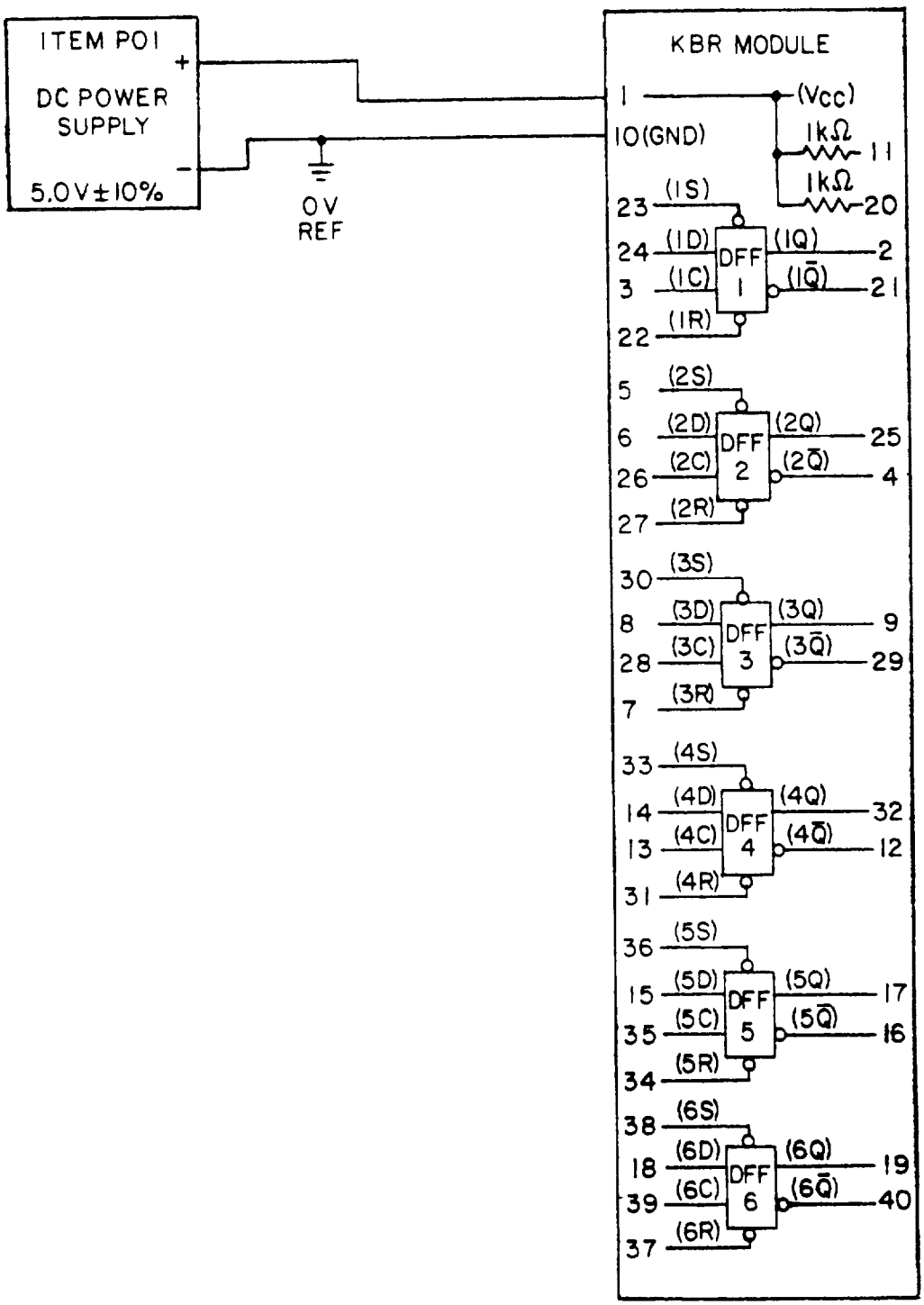


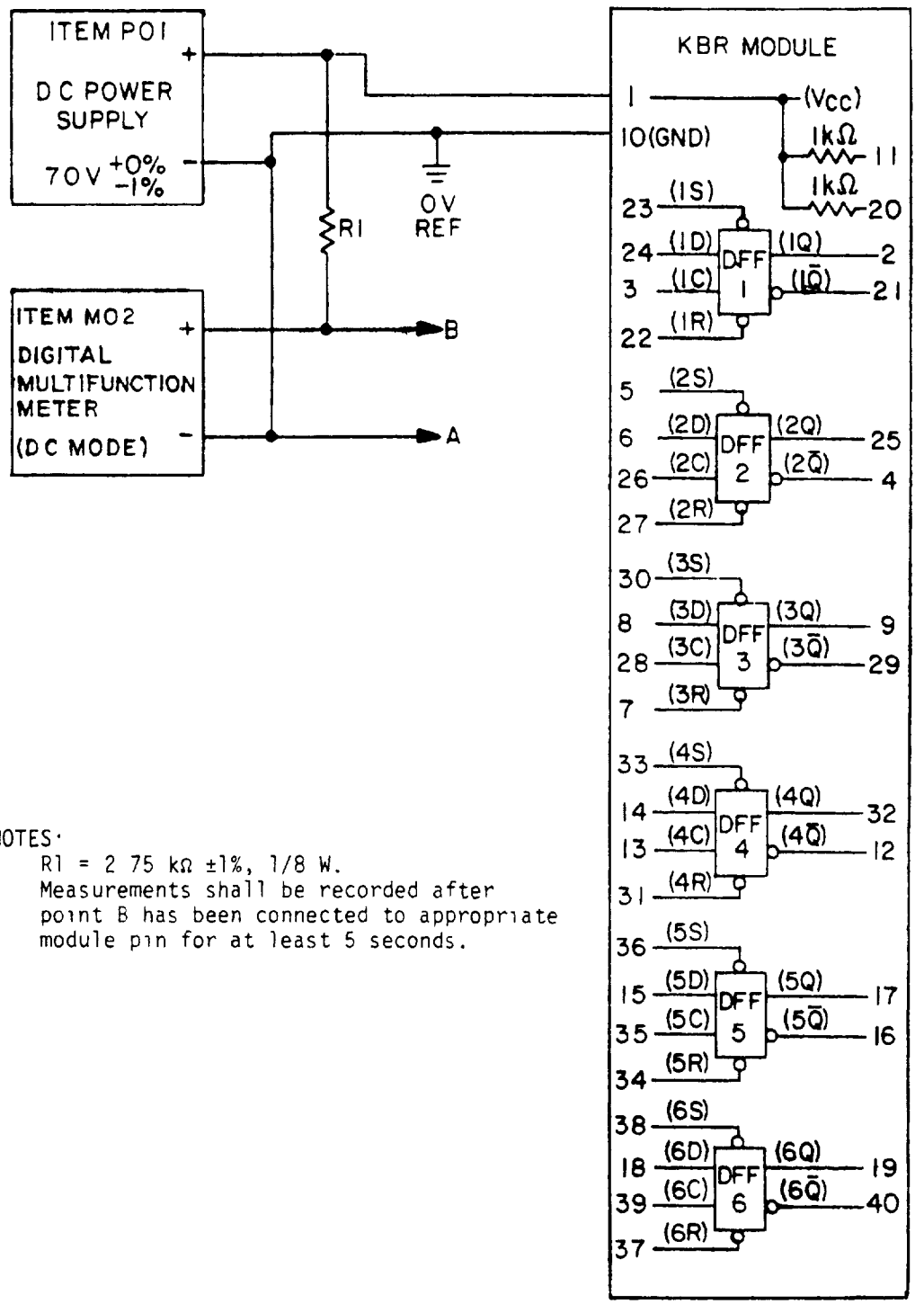
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1Q V_{OUT} (low loaded)	22	2	0.01	0.6	V	5
A2	$1\overline{Q}$	23	21				
A3	2Q	27	25				
A4	$2\overline{Q}$	5	4				
A5	3Q	7	9				
A6	$3\overline{Q}$	30	29				
A7	4Q	31	32				
A8	$4\overline{Q}$	33	12				
A9	5Q	34	17				
A10	$5\overline{Q}$	36	16				
A11	6Q	37	19				
A12	$6\overline{Q}$	38	40				

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NOTES:
1. R1 = 2 75 kΩ ±1%, 1/8 W.
2. Measurements shall be recorded after point B has been connected to appropriate module pin for at least 5 seconds.

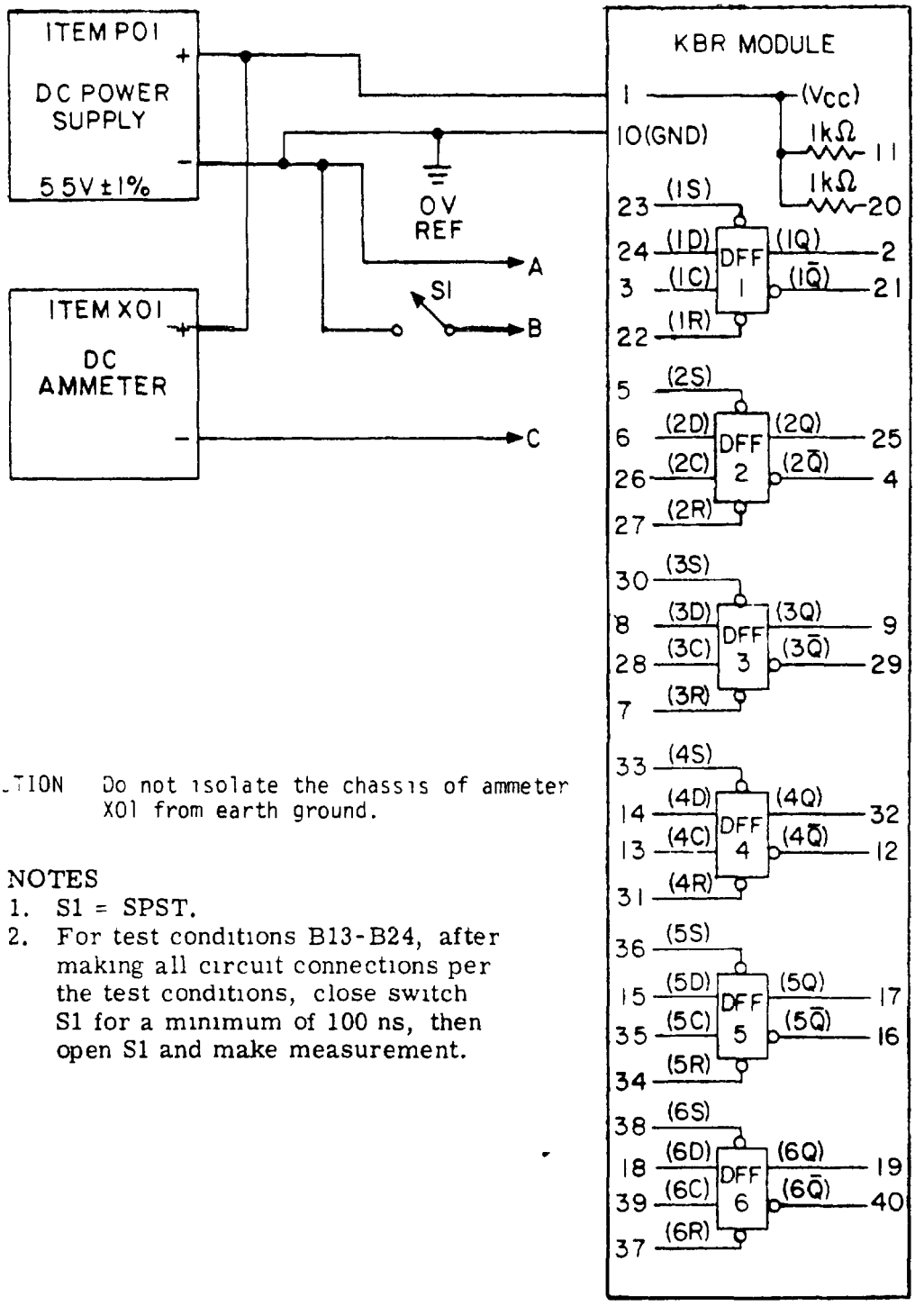
FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	1D Input breakdown current, data	22	Open	24	0.0	0.1	mA	6
B2	2D	27		6				
B3	3D	7		8				
B4	4D	31		14				
B5	5D	34		15				
B6	6D	37		18				
B7	1C Input breakdown current, clock	22		3		0.2		
B8	2C	27		26				
B9	3C	7		28				
B10	4C	31		13				
B11	5C	34		35				
B12	6C	37		39				
B13	1S Input breakdown current, set	Open	3	23				
B14	2S		26	5				
B15	3S		28	30				
B16	4S		13	33				
B17	5S		35	36				
B18	6S		39	38				
B19	1R Input breakdown current, reset	24	3	22		0.3		
B20	2R	6	26	27				
B21	3R	8	28	7				
B22	4R	14	13	31				
B23	5R	15	35	34				
B24	6R	18	39	37				

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CAUTION Do not isolate the chassis of ammeter X01 from earth ground.

NOTES
1. S1 = SPST.
2. For test conditions B13-B24, after making all circuit connections per the test conditions, close switch S1 for a minimum of 100 ns, then open S1 and make measurement.

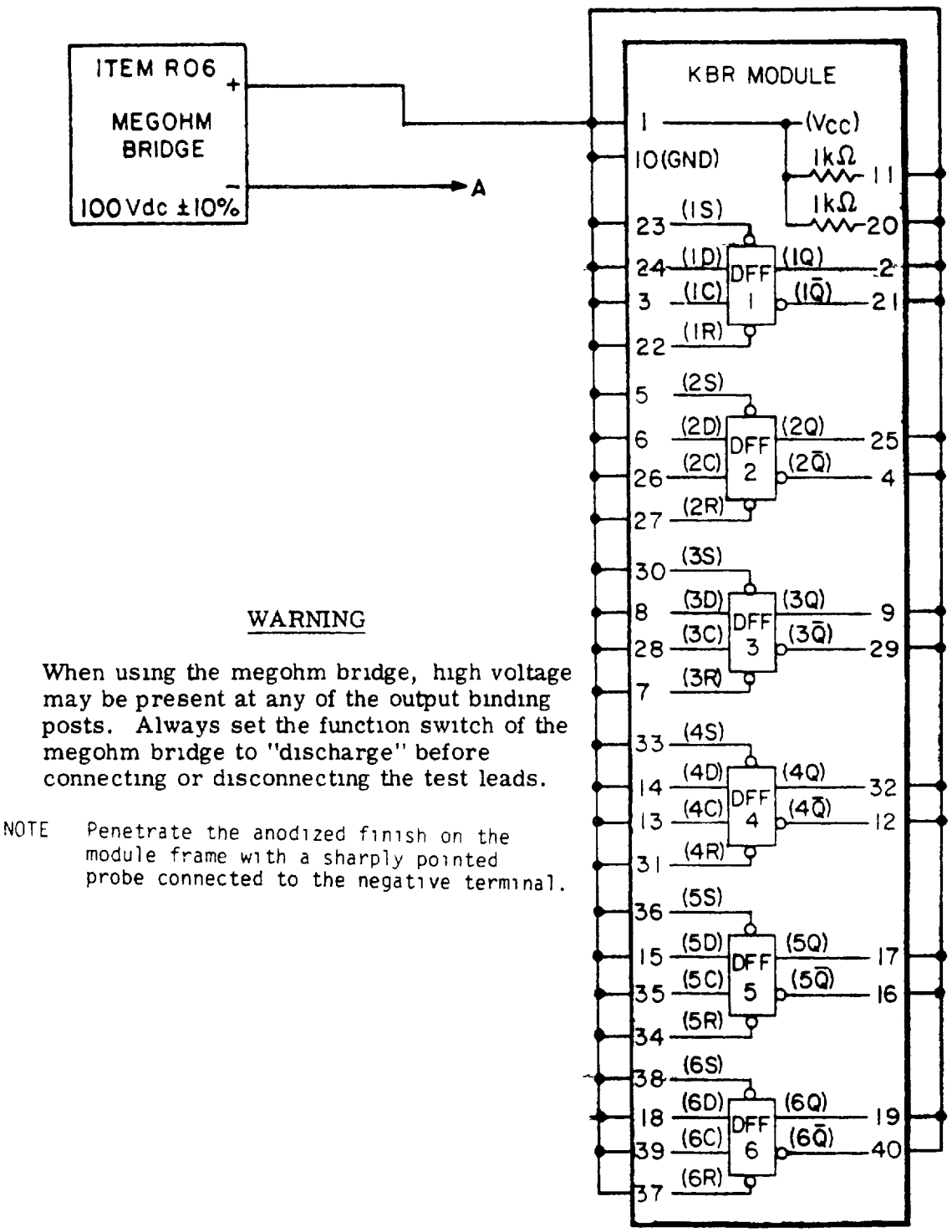
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C	Isolation	Frame	10	∞	MΩ	7

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

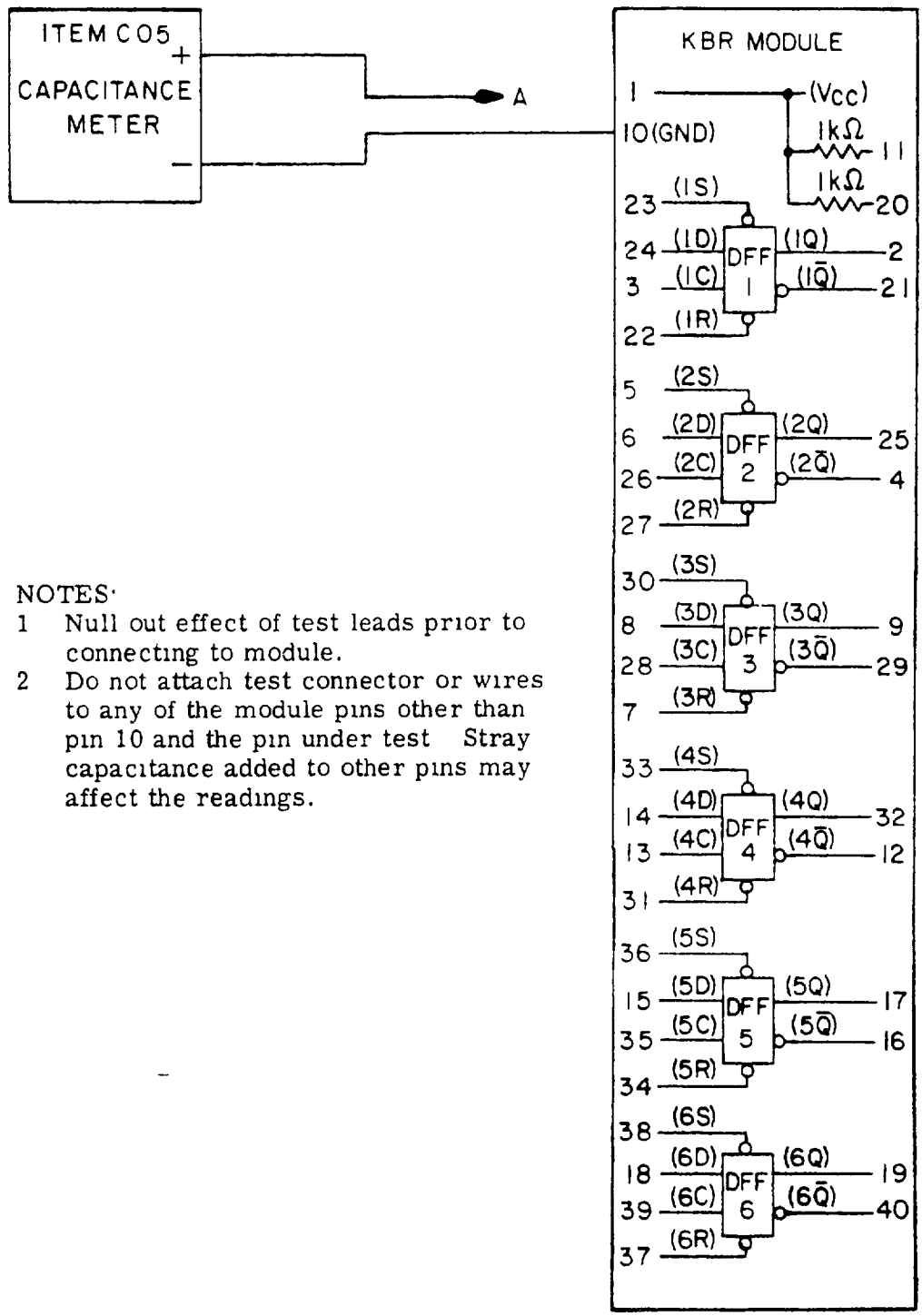
FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1C Input capacitance	3	2	15	pF	8
D2	1S	23				
D3	1R	22				
D4	1D	24				
D5	2S	5				
D6	2D	6				
D7	2R	27				
D8	2C	26				
D9	3D	8				
D10	3C	28				
D11	3S	30				
D12	3R	7				
D13	4C	13				
D14	4D	14				
D15	4S	33				
D16	4R	31				
D17	5D	15				
D18	5C	35				
D19	5S	36				
D20	5R	34				
D21	6D	18				
D22	6S	38				
D23	6C	39				
D24	6R	37				

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NOTES:
1. Null out effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 8. Input capacitance test circuit for table VI.

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TABLE VII. Static output voltage (low-level and high-level) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	1Q Static output voltage (threshold reset), low level	23,24	22	2	21	3	0.0	0.3	V	9
E2	2Q	5,6	27	25	4	26				
E3	3Q	8,30	7	9	29	28				
E4	4Q	14,33	31	32	12	13				
E5	5Q	15,36	34	17	16	35				
E6	6Q	18,38	37	19	40	39				
E7	$1\overline{Q}$ Static output voltage (threshold set), low level	22	23,24	21	2	3				
E8	$2\overline{Q}$	27	5,6	4	25	26				
E9	$3\overline{Q}$	7	8,30	29	9	28				
E10	$4\overline{Q}$	31	14,33	12	32	13				
E11	$5\overline{Q}$	34	15,36	16	17	35				
E12	$6\overline{Q}$	37	18,38	40	19	39				
E13	1Q Static output voltage (threshold D), low level	22,23	24	2	21	3				
E14	2Q	5,27	6	25	4	26				
E15	3Q	7,30	8	9	29	28				
E16	4Q	31,33	14	32	12	13				
E17	5Q	34,36	15	17	16	35				
E18	6Q	37,38	18	19	40	39				
E19	$1\overline{Q}$	22,23, 24	Open	21	2	3				
E20	$2\overline{Q}$	5,6,27	Open	4	25	26				

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TABLE VII. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E21	$3\overline{Q}$ Static output voltage (threshold D), low level	7,8,30	Open	29	9	28	0.0	0.3	V	9
E22	$4\overline{Q}$	14,31, 33		12	32	13				
E23	$5\overline{Q}$	15,34, 36		16	17	35				
E24	$6\overline{Q}$	18,37, 38		40	19	39				
F1	1Q Static output voltage (threshold set), high level	22	23,24	2	21	3	2.4	4.5		
F2	2Q	27	5,6	25	4	26				
F3	3Q	7	8,30	9	29	28				
F4	4Q	31	14,33	32	12	13				
F5	5Q	34	15,36	17	16	35				
F6	6Q	37	18,38	19	40	39				
F7	$1\overline{Q}$ Static output voltage (threshold reset), high level	23,24	22	21	2	3				
F8	$2\overline{Q}$	5,6	27	4	25	26				
F9	$3\overline{Q}$	8,30	7	29	9	28				
F10	$4\overline{Q}$	14,33	31	12	32	13				
F11	$5\overline{Q}$	15,36	34	16	17	35				
F12	$6\overline{Q}$	18,38	37	40	19	39				
F13	1Q Static output voltage (threshold D), high level	22,23, 24	Open	2	21	3				
F14	2Q Static output voltage (threshold D), high level	5,6,27	Open	25	4	26				

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TABLE VII. Static output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F15	3Q Static output voltage (threshold D), high level	7,8,30	Open	9	29	28	2.4	4.5	V	9
F16	4Q	14,31, 33		32	12	13				
F17	5Q	15,34, 36		17	16	35				
F18	6Q	18,37, 38		19	40	39				
F19	$1\overline{Q}$	22,23	24	21	2	3				
F20	$2\overline{Q}$	5,27	6	4	25	26				
F21	$3\overline{Q}$	7,30	8	29	9	28				
F22	$4\overline{Q}$	31,33	14	12	32	13				
F23	$5\overline{Q}$	34,36	15	16	17	35				
F24	$6\overline{Q}$	37,38	18	40	19	39				
F25	1Q Static output voltage (threshold set-reset), high level	Open	22,23, 24	2	21	3				
F26	2Q		5,6,27	25	4	26				
F27	3Q		7,8,30	9	29	28				
F28	4Q		14,31, 33	32	12	13				
F29	5Q		15,34, 36	17	16	35				
F30	6Q		18,37, 38	19	40	39				
F31	$1\overline{Q}$	24	22,23	21	2	3				
F32	$2\overline{Q}$	6	5,27	4	25	26				

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TABLE VII. Static otuput voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F33	$3\overline{Q}$ Static output voltage (threshold set-reset), high level	8	7,30	29	9	28	2.4	4.5	V	9
F34	$4\overline{Q}$	14	31,33	12	32	13	↓	↓	↓	↓
F35	$5\overline{Q}$	15	34,36	16	17	35				
F36	$6\overline{Q}$ ↓	18	37,38	40	19	39				

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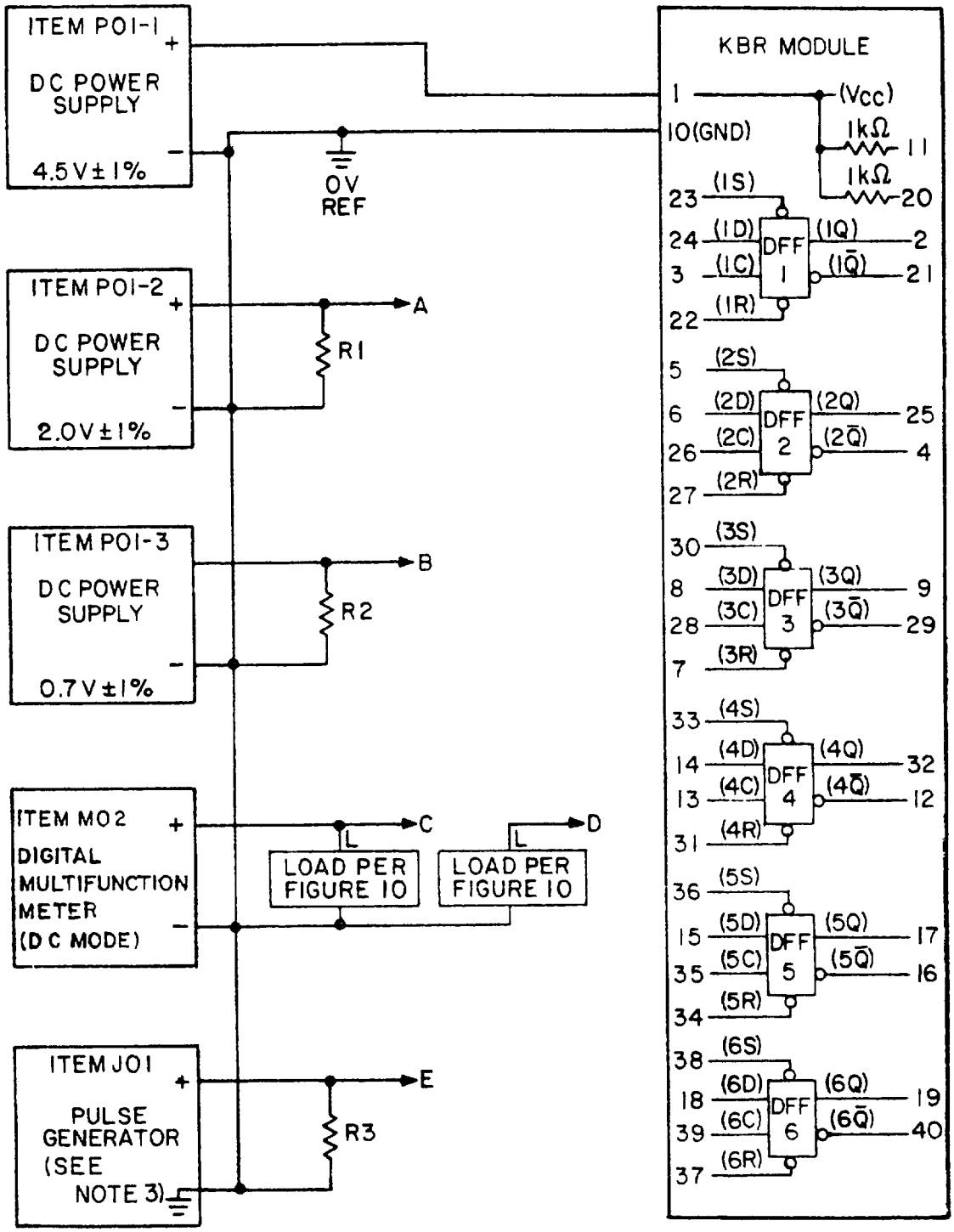
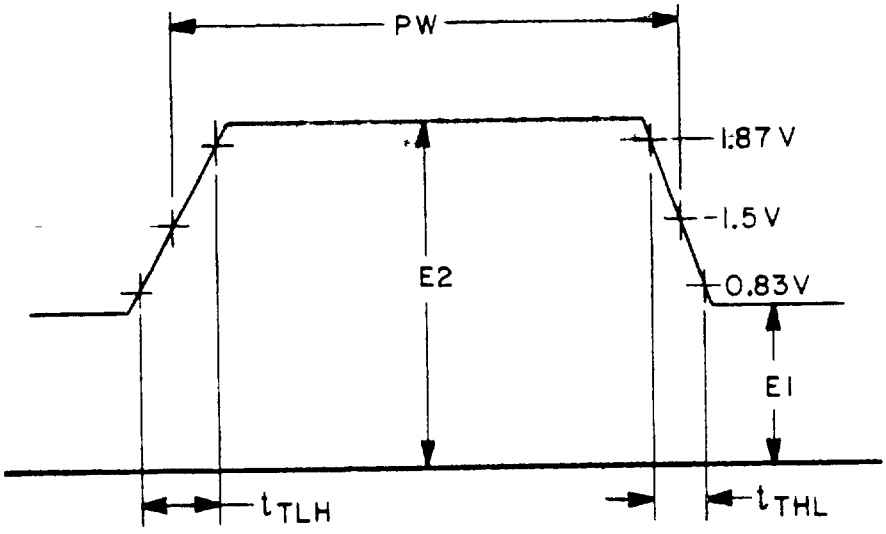


FIGURE 9. Static output voltage (low-level and high-level) test circuit for table VII.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1 W.
2. After all circuit connections are made per table VII for each test condition, apply one pulse with item J01 and make measurement.

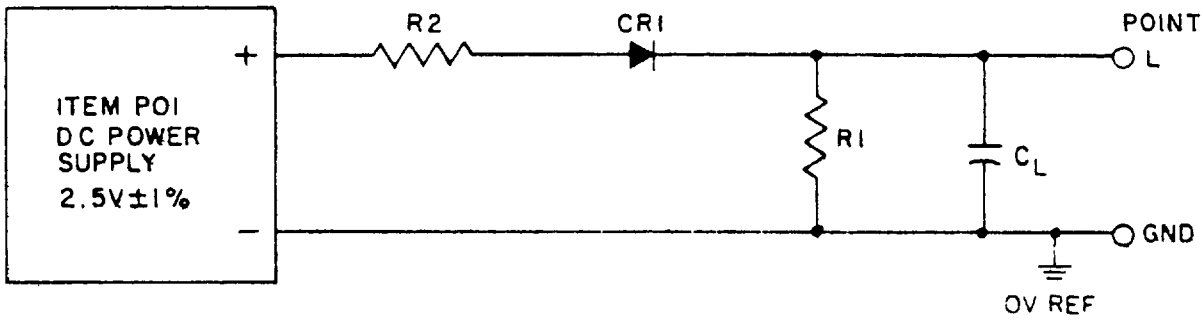
3. Item J01 setup.



(a) PW = 500 ±50 ns
(b) E1 = 0.7 ±0.05 V.
(c) E2 = 2.0 ±0.05 V.
(d) t_{TLH} = 25 ±20 ns.
(e) t_{THL} = 25 ±20 ns.
(f) For pulse setup, use a repetition rate of 1.0 MHz ±20%.
(g) After setting up the pulse, place item J01 in the one-shot mode

FIGURE 9. Static output voltage (low-level and high-level) test circuit for table VII - Continued.

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NOTES
1. For test conditions E1-E24, F1-F36, L1-L12, M1-M12, N1-N12, P1-P12, V1-V12, and W1-W12
   (a) C_L = 150 ±5 pF (including probe and parasitic capacitance)
   (b) CR1 = 1N4150 or equivalent
   (c) R1 = 24 3 kΩ ±1%, 1/8 W (current flow of 100 µA when point L is held at 2 4 V)
   (d) R2 is selected for current flow of 2 0 mA ±2% out of point L when point L is held at 0.3 V
2. For test conditions L13-L24, M13-M24, N13-N24, P13-P24, R1-R12, S1-S12, T1-T24, V13-V24, and W13-W24.
   (a) C_L = 50 ±5 pF (including probe and parasitic capacitance).
   (b) Same as 1(b)
   (c) Same as 1(c).
   (d) Same as 1(d)
3. Capacitor C_L is not required for test conditions E1-E24 and F1-F36
4. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76 20 mm)
5. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV

FIGURE 10. Logic load circuit.

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TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1D Input current (data), low level	3,22	Open	23	24	0.05	0.20	mA	11
G2	2D	26,27		5	6				
G3	3D	7,28		30	8				
G4	4D	13,31		33	14				
G5	5D	34,35		36	15				
G6	6D	37,39		38	18				
G7	1S Input current (set), low level	22		3,24	23		0.40		
G8	2S	27		6,26	5				
G9	3S	7		8,28	30				
G10	4S	31		13,14	33				
G11	5S	34		15,35	36				
G12	6S	37		18,39	38				
G13	1C Input current(clock), low level	22		23,24	3				
G14	2C	27		5,6	26				
G15	3C	7		8,30	28				
G16	4C	31		14,33	13				
G17	5C	34		15,36	35				
G18	6C	37		18,38	39				
G19	1R Input current(reset), low level	3,23,24		Open	22		0.60		
G20	2R	5,6,26			27				
G21	3R	8,28,30			7				
G22	4R	13,14,33			31				
G23	5R	15,35,36			34				
G24	6R	18,38,39			37				

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	E to pin	Min	Max		
H1	1D Input current (data), high level	3,23	24	22	Open	0.0	10.0	µA	11
H2	2D	5,26	6	27					
H3	3D	28,30	8	7					
H4	4D	13,33	14	31					
H5	5D	35,36	15	34					
H6	6D	38,39	18	37					
H7	1C Input current (clock), high level	23,24	3	22			20.0		
H8	2C	5,6	26	27					
H9	3C	8,30	28	7					
H10	4C	14,33	13	31					
H11	5C	15,36	35	34					
H12	6C	18,38	39	37					
H13	1C	22,24	3	23					
H14	2C	6,27	26	5					
H15	3C	7,8	28	30					
H16	4C	14,31	13	33					
H17	5C	15,34	35	36					
H18	6C	18,37	39	38					
H19	1S Input current (set), high level	22,24	23	Open	3				
H20	2S	6,27	5		26				
H21	3S	7,8	30		28				
H22	4S	14,31	33		13				

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TABLE IX. Input current (high-level) test - Continued.

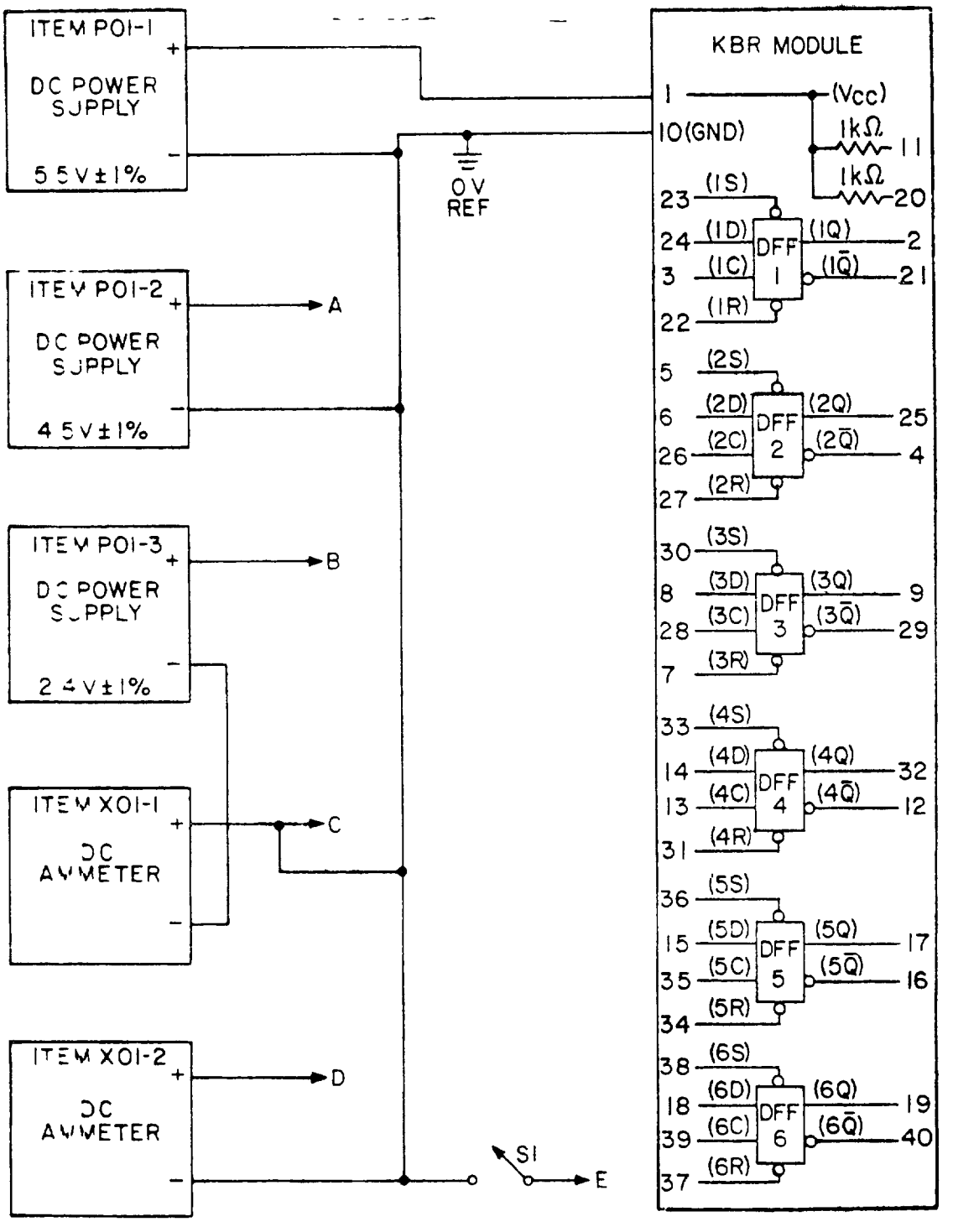
Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	P to pin	C to pin	I to pin	Min	Max		
H23	5S Input current (set), high level	15,34	36	Open	35	0.0	20.0	µA	11
H24	6S Input current (set), high level	18,37	38	Open	39		20.0		
H25	1R Input current (reset), high level	23	22	24	3		30.0		
H26	2R	5	27	6	26				
H27	3R	30	7	8	28				
H28	4R	33	31	14	13				
H29	5R	36	34	15	35				
H30	6R	38	37	18	39				

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TABLE X. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		C to pin	D to pin	Min	Max		
J1	1Q Output short circuit current	23	2	3.0	15.0	mA	11
J2	$1\overline{Q}$	22	21				
J3	2Q	5	25				
J4	$2\overline{Q}$	27	4				
J5	3Q	30	9				
J6	$3\overline{Q}$	7	29				
J7	4Q	33	32				
J8	$4\overline{Q}$	31	12				
J9	5Q	36	17				
J10	$5\overline{Q}$	34	16				
J11	6Q	38	19				
J12	$6\overline{Q}$	37	40	3.0	15.0		
J13		Open	11	5.1	5.9		
J14		Open	20	5.1	5.9		

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CAUTION Do not isolate the chassis of ammeter XO1 from earth ground

FIGURE 11 Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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NOTES:
1. S1 = SPST.
2. Since items XO1-1 and XO1-2 are not used simultaneously, only one ammeter is required. Use XO1-2 for tables VIII and X; use item XO1-1 for table IX.
3. With point B disconnected, item XO1-1 should indicate zero current on the 1 µA range.
4. For test conditions H19-H30, after making all circuit connections per the test conditions, close switch S1 for a minimum of 100 ns, then open S1 and make measurement.
5. When performing output short circuit current tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

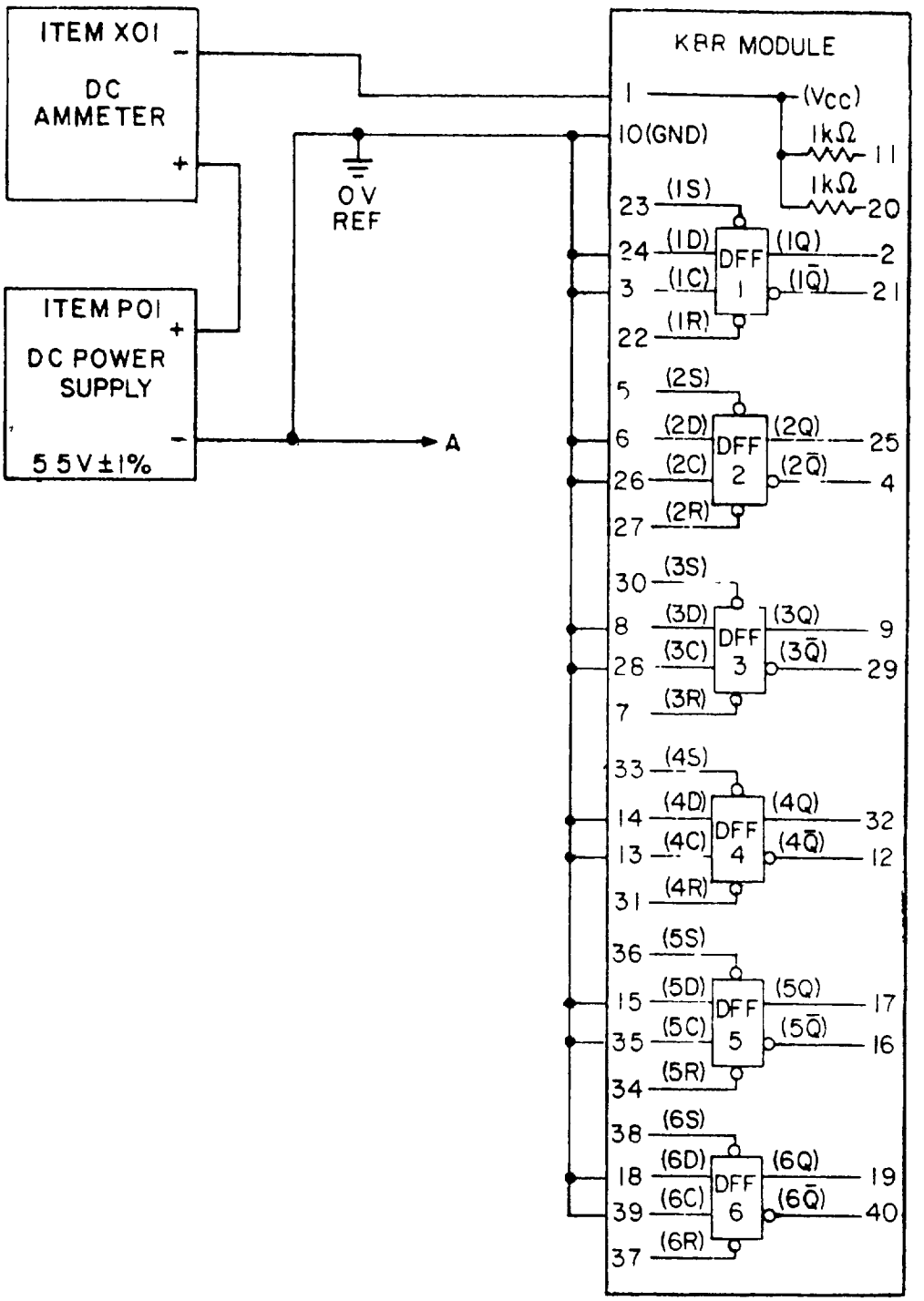
FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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TABLE XI. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply current, high level	23,5,30,33,36,38	1.0	9.0	mA	12
K2	Power supply current, low level	22,27,7,31,34,37	1.0	9.0	mA	12

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CAUTION Do not isolate the chassis of ammeter XOI from earth ground

FIGURE 12. Power supply current (high-level and low-level) test circuit for table XI.

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TABLE XII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1Q Transition time (t_{TLH}), C_L = 150 pF	3	24	2	21	10	100	ns	13,14
L2	$1\overline{Q}$	3	24	21	2				
L3	2Q	26	6	25	4				
L4	$2\overline{Q}$	26	6	4	25				
L5	3Q	28	8	9	29				
L6	$3\overline{Q}$	28	8	29	9				
L7	4Q	13	14	32	12				
L8	$4\overline{Q}$	13	14	12	32				
L9	5Q	35	15	17	16				
L10	$5\overline{Q}$	35	15	16	17				
L11	6Q	39	18	19	40				
L12	$6\overline{Q}$	39	18	40	19				
L13	1Q Transition time (t_{TLH}), C_L = 50 pF	3	24	2	21	5	60		
L14	$1\overline{Q}$	3	24	21	2				
L15	2Q	26	6	25	4				
L16	$2\overline{Q}$	26	6	4	25				
L17	3Q	28	8	9	29				
L18	$3\overline{Q}$	28	8	29	9				
L19	4Q	13	14	32	12				
L20	$4\overline{Q}$	13	14	12	32				
L21	5Q	35	15	17	16				
L22	$5\overline{Q}$	35	15	16	17				

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TABLE XII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L23	6Q Transition time (t_{TLH}), C_L = 50 pF	39	18	19	40	5	60	ns	13,14
L24	6$\overline{Q}$ Transition time (t_{TLH}), C_L = 50 pF	39	18	40	19	5	60	ns	13,14

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Q Transition time (t_{THL}), C_L = 150 pF	3	24	2	21	5	60	ns	13,14
M2	$1\overline{Q}$	3	24	21	2				
M3	2Q	26	6	25	4				
M4	$2\overline{Q}$	26	6	4	25				
M5	3Q	28	8	9	29				
M6	$3\overline{Q}$	28	8	29	9				
M7	4Q	13	14	32	12				
M8	$4\overline{Q}$	13	14	12	32				
M9	5Q	35	15	17	16				
M10	$5\overline{Q}$	35	15	16	17				
M11	6Q	39	18	19	40				
M12	$6\overline{Q}$	39	18	40	19				
M13	1Q Transition time (t_{THL}), C_L = 50 pF	3	24	2	21	3	30		
M14	$1\overline{Q}$	3	24	21	2				
M15	2Q	26	6	25	4				
M16	$2\overline{Q}$	26	6	4	25				
M17	3Q	28	8	9	29				
M18	$3\overline{Q}$	28	8	29	9				
M19	4Q	13	14	32	12				
M20	$4\overline{Q}$	13	14	12	32				
M21	5Q	35	15	17	16				
M22	$5\overline{Q}$	35	15	16	17				
M23	6Q	39	18	19	40				
M24	$6\overline{Q}$	39	18	40	19				

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TABLE XIV. Propagation delay time (t_{PHL})(clock-to-output) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Q Propagation delay time (t_{PHL}), clock to output, C_L = 150 pF	3	24	2	21	25	200	ns	13,14
N2	$1\overline{Q}$	3	24	21	2				
N3	2Q	26	6	25	4				
N4	$2\overline{Q}$	26	6	4	25				
N5	3Q	28	8	9	29				
N6	$3\overline{Q}$	28	8	29	9				
N7	4Q	13	14	32	12				
N8	$4\overline{Q}$	13	14	12	32				
N9	5Q	35	15	17	16				
N10	$5\overline{Q}$	35	15	16	17				
N11	6Q	39	18	19	40				
N12	$6\overline{Q}$	39	18	40	19				
N13	1Q Propagation delay time (t_{PHL}), clock to output, C_L = 50 pF	3	24	2	21		150		
N14	$1\overline{Q}$	3	24	21	2				
N15	2Q	26	6	25	4				
N16	$2\overline{Q}$	26	6	4	25				
N17	3Q	28	8	9	29				
N18	$3\overline{Q}$	28	8	29	9				
N19	4Q	13	14	32	12				
N20	$4\overline{Q}$	13	14	12	32				
N21	5Q	35	15	17	16				
N22	$5\overline{Q}$	35	15	16	17				

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TABLE XIV. Propagation delay time (t_{PHL})(clock-to-output) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N23	6Q Propagation delay time (t_{PHL}), clock to output, C_L = 50 pF	39	18	19	40	25	150	ns	13,14
N24	6$\overline{Q}$ Propagation delay time (t_{PHL}), clock to output, C_L = 50 pF	39	18	40	19	25	150	ns	13,14

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TABLE XV. Propagation delay time (t_{PLH})(clock-to-output) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Q Propagation delay time (t_{PLH}), clock to output, C_L = 150 pF	3	24	2	21	25	140	ns	13,14
P2	$1\overline{Q}$	3	24	21	2				
P3	2Q	26	6	25	4				
P4	$2\overline{Q}$	26	6	4	25				
P5	3Q	28	8	9	29				
P6	$3\overline{Q}$	28	8	29	9				
P7	4Q	13	14	32	12				
P8	$4\overline{Q}$	13	14	12	32				
P9	5Q	35	15	17	16				
P10	$5\overline{Q}$	35	15	16	17				
P11	6Q	39	18	19	40				
P12	$6\overline{Q}$	39	18	40	19				
P13	1Q Propagation delay time (t_{PLH}), clock to output, C_L = 50 pF	3	24	2	21		100		
P14	$1\overline{Q}$	3	24	21	2				
P15	2Q	26	6	25	4				
P16	$2\overline{Q}$	26	6	4	25				
P17	3Q	28	8	9	29				
P18	$3\overline{Q}$	28	8	29	9				
P19	4Q	13	14	32	12				
P20	$4\overline{Q}$	13	14	12	32				
P21	5Q	35	15	17	16				
P22	$5\overline{Q}$	35	15	16	17				

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TABLE XV. Propagation delay time (t_{PLH})(clock-to-output) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P23	6Q Propagation delay time (t_{PLH}), clock to output, C_L = 50 pF	39	18	19	40	25	100	ns	13,14
P24	6$\overline{Q}$ Propagation delay time (t_{PLH}), clock to output, C_L = 50 pF	39	18	40	19	25	100	ns	13,14

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TABLE XVI. Data setup time (maximum clock frequency, minimum clock pulse width) check test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q Data setup time (maximum clock frequency, minimum clock pulse width) check	3	24	2	21	780	840	ns	13,14
R2	$1\overline{Q}$	3	24	21	2				
R3	2Q	26	6	25	4				
R4	$2\overline{Q}$	26	6	4	25				
R5	3Q	28	8	9	29				
R6	$3\overline{Q}$	28	8	29	9				
R7	4Q	13	14	32	12				
R8	$4\overline{Q}$	13	14	12	32				
R9	5Q	35	15	17	16				
R10	$5\overline{Q}$	35	15	16	17				
R11	6Q	39	18	19	40				
R12	$6\overline{Q}$	39	18	40	19				

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TABLE XVII. Data hold time check test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Q Data hold time check	3	24	2	21	780	840	ns	13,14
S2	$1\overline{Q}$	3	24	21	2				
S3	2Q	26	6	25	4				
S4	$2\overline{Q}$	26	6	4	25				
S5	3Q	28	8	9	29				
S6	$3\overline{Q}$	28	8	29	9				
S7	4Q	13	14	32	12				
S8	$4\overline{Q}$	13	14	12	32				
S9	5Q	35	15	17	16				
S10	$5\overline{Q}$	35	15	16	17				
S11	6Q	39	18	19	40				
S12	$6\overline{Q}$	39	18	40	19				

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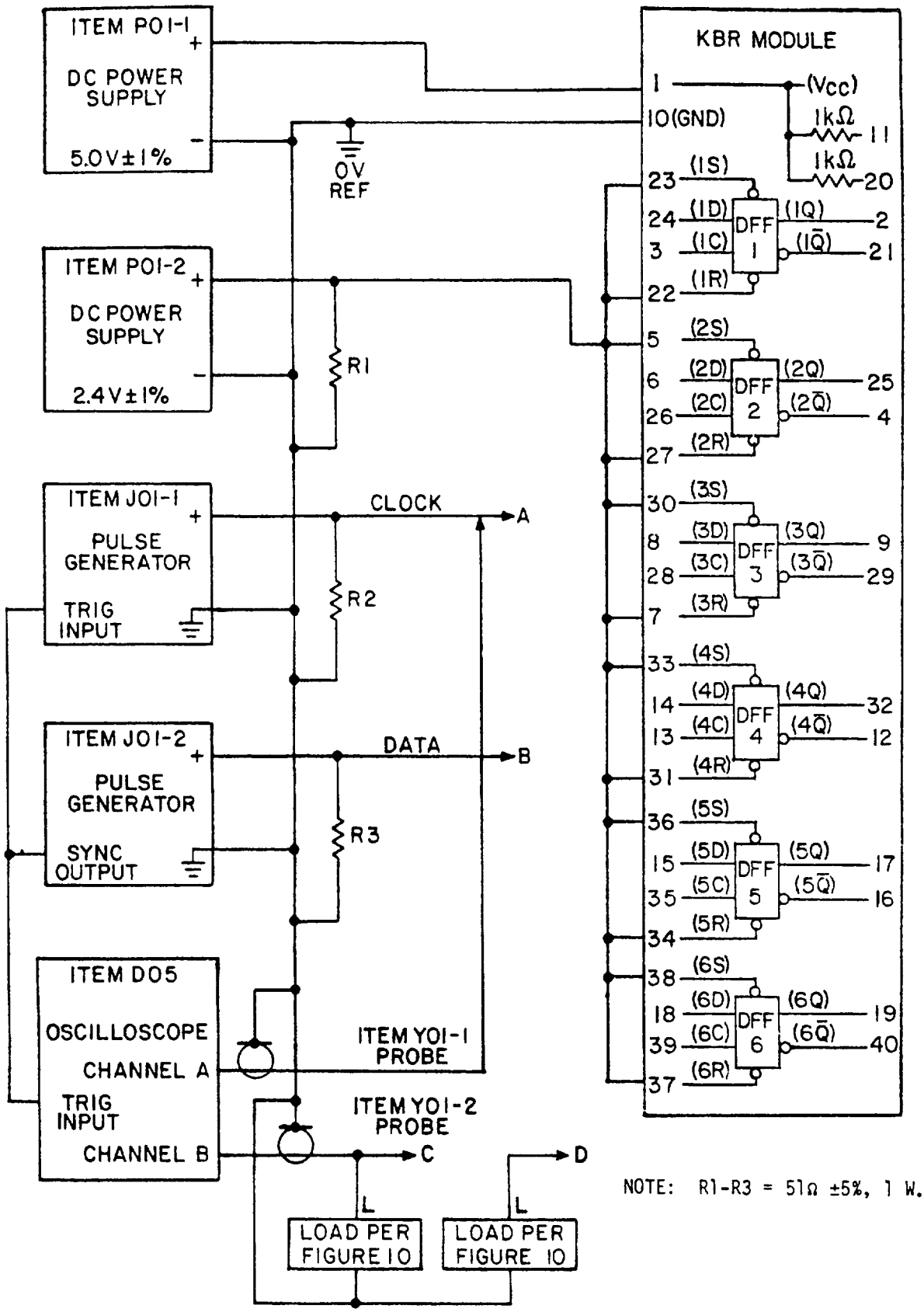


FIGURE 13. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} and t_{PLH}) (clock-to-output), data setup time (maximum clock frequency, minimum pulse width) check, and data hold time check test circuit for tables XII, XIII, XIV, XV, XVI, and XVII.

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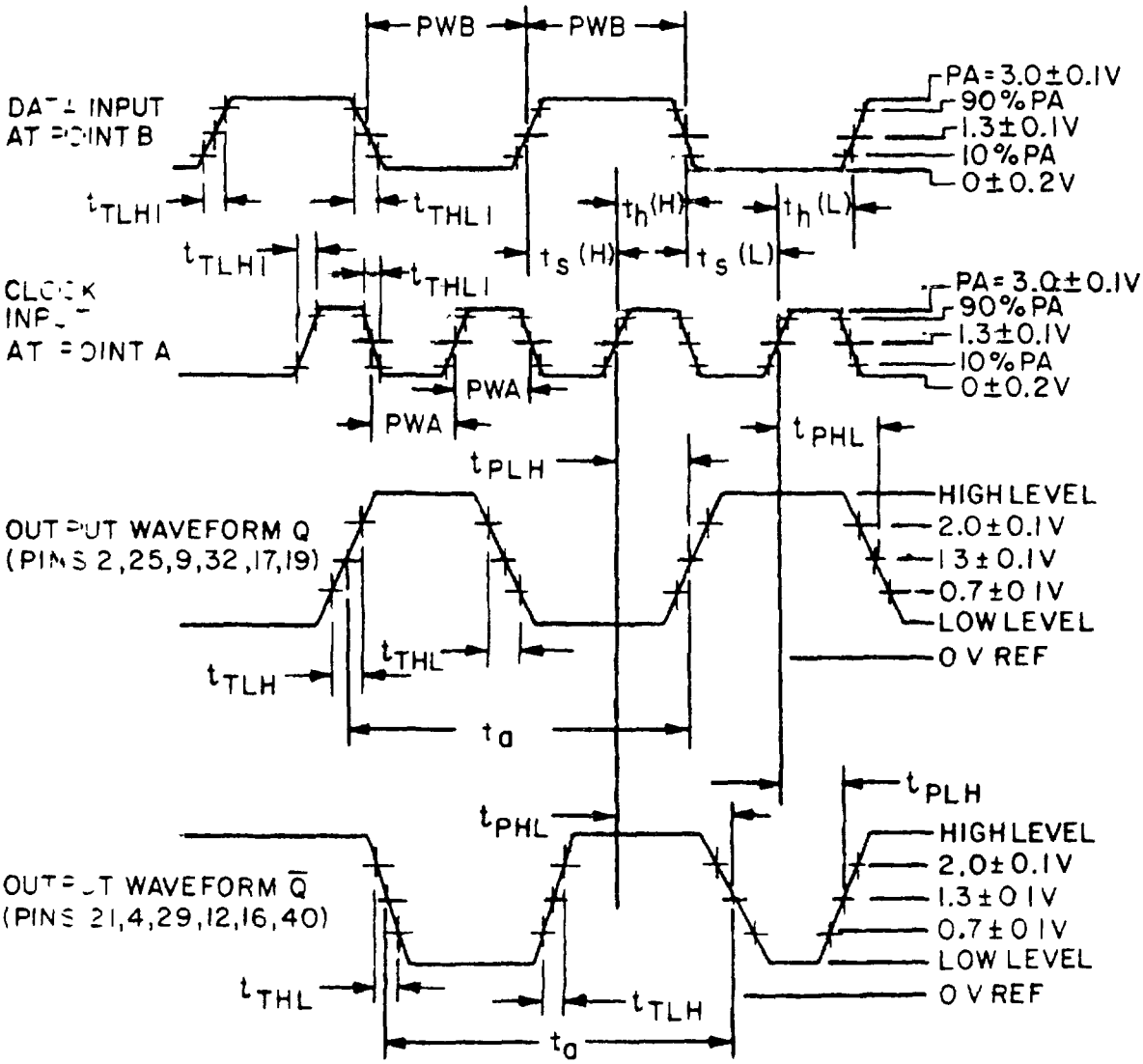


FIGURE 14. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} and t_{PLH}) (clock-to-output), data setup time (maximum clock frequency, minimum clock pulse width) check, and data hold time check test circuit waveforms for tables XII, XIII, XIV, XV, XVI, and XVII

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NOTES:

1. Input pulse parameters:
   (a) PWB = Data pulse width.
   (b) PWA = Clock pulse width.
   (c) t_h = Hold time.
   (d) t_s = Setup time.
   (e) t_a = Test parameter.
   (f) t_{TLH} and t_{THL} = Output transition time.
   (g) t_{TLH_1} = t_{THL_1} = Input transition time.
2. Test conditions L1-L24, M1-M24, N1-N24, and P1-P24
   (a) Adjust items J01-1 and J01-2 to provide input pulse parameters as follows
      (1) t_{TLH_1} = and t_{THL_1} = 10 ±2 ns
      (2) PWD = 500 ns ±10%.
      (3) PWA = 250 ns ±10%.
   (b) $t_s(H)$ = 125 ns ±10%.
3. Test conditions R1-R12
   (a) Adjust items J01-1 and J01-2 to provide input pulse parameters as follows
      (1) t_{TLH_1} and t_{THL_1} = 10 ±2 ns.
      (2) PWA = 200 $^{+10}_{-0}$ ns.
      (3) PWB = 400 ±20 ns.
   (b) $t_s(H) = t_s(L)$ = 50 $^{+2}_{-0}$ ns.
4. Test conditions S1-S12.
   (a) Same as note 3(a) above.
   (b) $t_h(H) = t_h(L)$ = 25 $^{+2}_{-0}$ ns.
5. When performing any of the above tests, the high level pulse must be verified to be greater than 2.4 V, and the low level must be verified to be less than 0.3 V.
6. All voltage levels are referenced to ground.

FIGURE 14. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} and t_{PLH}) (clock-to-output), data setup time (maximum clock frequency, minimum clock pulse width) check, and data hold time check test circuit waveforms for tables XII, XIII, XIV, XV, XVI, and XVII - Continued.

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TABLE XVIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1Q Output crosstalk, low level	23,27,5,7,30,31,33,34, 36,37,38	22	2	0	400	mV	15,16
T2	1$\overline{Q}$ Output crosstalk, high level	23,27,5,7,30,31,33,34, 36,37,38	22	21				
T3	1$\overline{Q}$ Output crosstalk, low level	22,27,5,7,30,31,33,34, 36,37,38	23	21				
T4	1Q Output crosstalk, high level	22,27,5,7,30,31,33,34, 36,37,38	23	2				
T5	2Q Output crosstalk, low level	22,23,5,7,30,31,33,34, 36,37,38	27	25				
T6	2$\overline{Q}$ Output crosstalk, high level	22,23,5,7,30,31,33,34, 36,37,38	27	4				
T7	2$\overline{Q}$ Output crosstalk, low level	22,23,27,7,30,31,33,34, 36,37,38	5	4				
T8	2Q Output crosstalk, high level	22,23,27,7,30,31,33,34, 36,37,38	5	25				
T9	3Q Output crosstalk, low level	22,23,27,5,30,31,33,34, 36,37,38	7	9				
T10	3$\overline{Q}$ Output crosstalk, high level	22,23,27,5,30,31,33,34, 36,37,38	7	29				
T11	3$\overline{Q}$ OUtput crosstalk, low level	22,23,27,5,7,31,33,34, 36,37,38	30	29				
T12	3Q Output crosstalk, high level	22,23,27,5,7,31,33,34, 36,37,38	30	9				
T13	4Q Output crosstalk, low level	22,23,27,5,7,30,33,34, 36,37,38	31	32				
T14	4$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,33,34, 36,37,38	31	12				
T15	4$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,34, 36,37,38	33	12				

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TABLE XVIII. Output crosstalk (low-level and high-level) test - Continued.

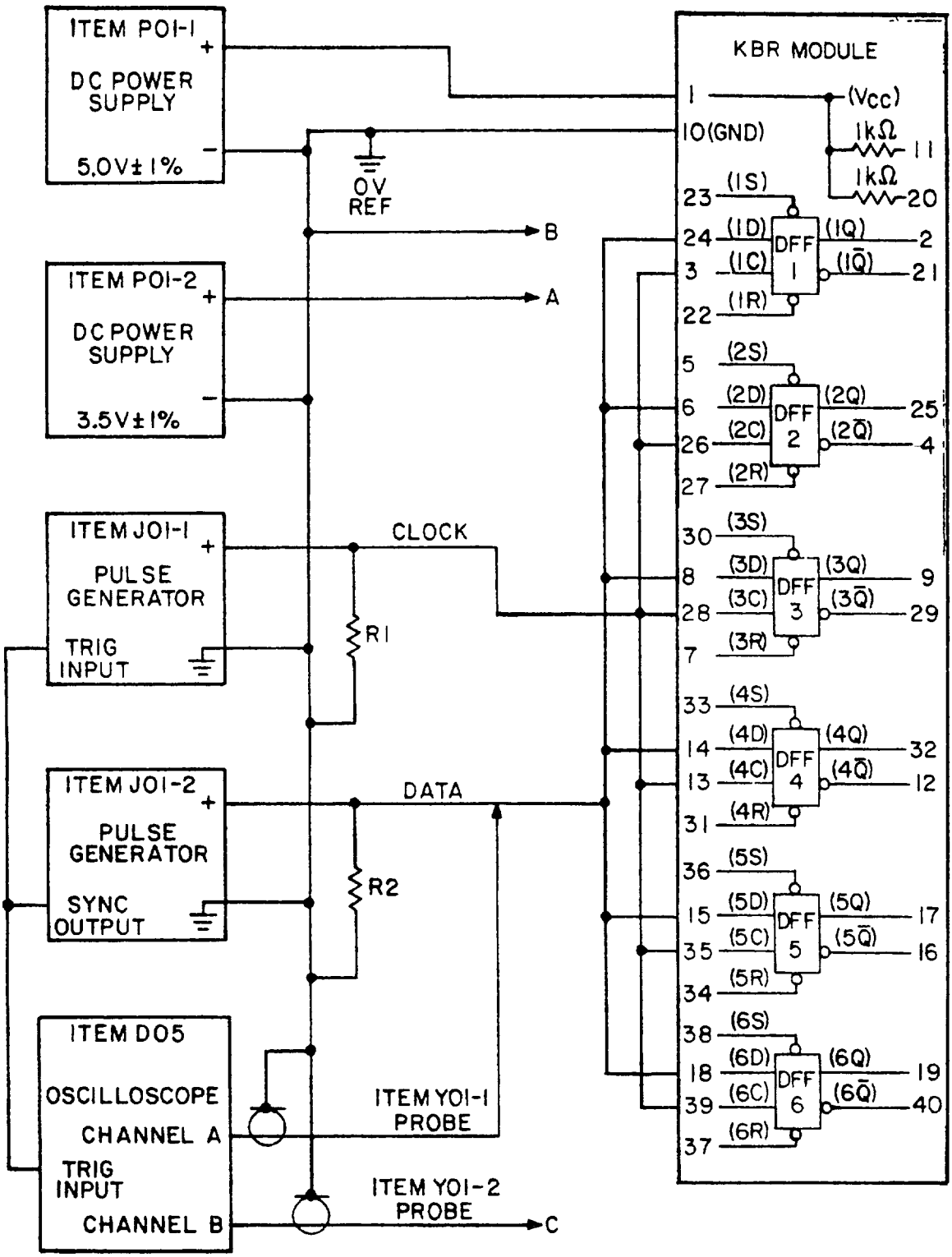
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T16	4Q Output crosstalk, high level	22,23,27,5,7,30,31,34, 36,37,38	33	32	0	400	mV	15,16
T17	5Q Output crosstalk, low level	22,23,27,5,7,30,31,33, 36,37,38	34	17				
T18	5$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,31,33, 36,37,38	34	16				
T19	5$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,33, 34,37,38	36	16				
T20	5Q Output crosstalk, high level	22,23,27,5,7,30,31,33, 34,37,38	36	17				
T21	6Q Output crosstalk, low level	22,23,27,5,7,30,31,33, 34,36,38	37	19				
T22	6$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,31,33, 34,36,38	37	40				
T23	6$\overline{Q}$ Output crosstalk, low level	22,23,27,5,7,30,31,33, 34,36,37	38	40				
T24	6$\overline{Q}$ Output crosstalk, high level	22,23,27,5,7,30,31,33, 34,36,37	38	19				

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U	Power supply transient amplitude	22,23,5,27,7,30,31,33, 34,36,37,38	Open	1	0	400	mV	15,17

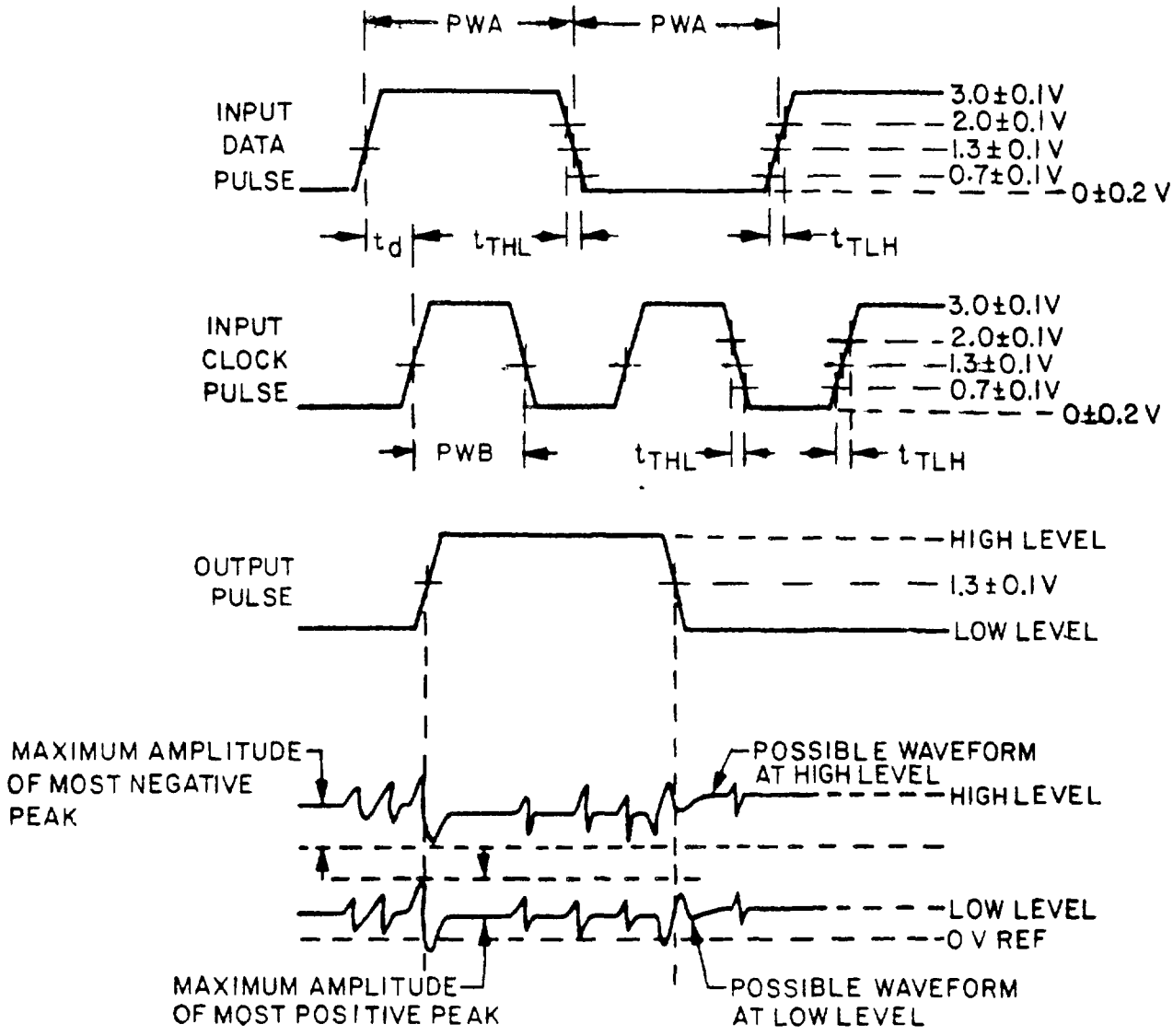
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. For test conditions T1-T24, load each output with a load as specified on figure 10.

FIGURE 15. Output crosstalk (low-level and high-level) and power supply transient amplitude test circuit for tables XVIII and XIX.

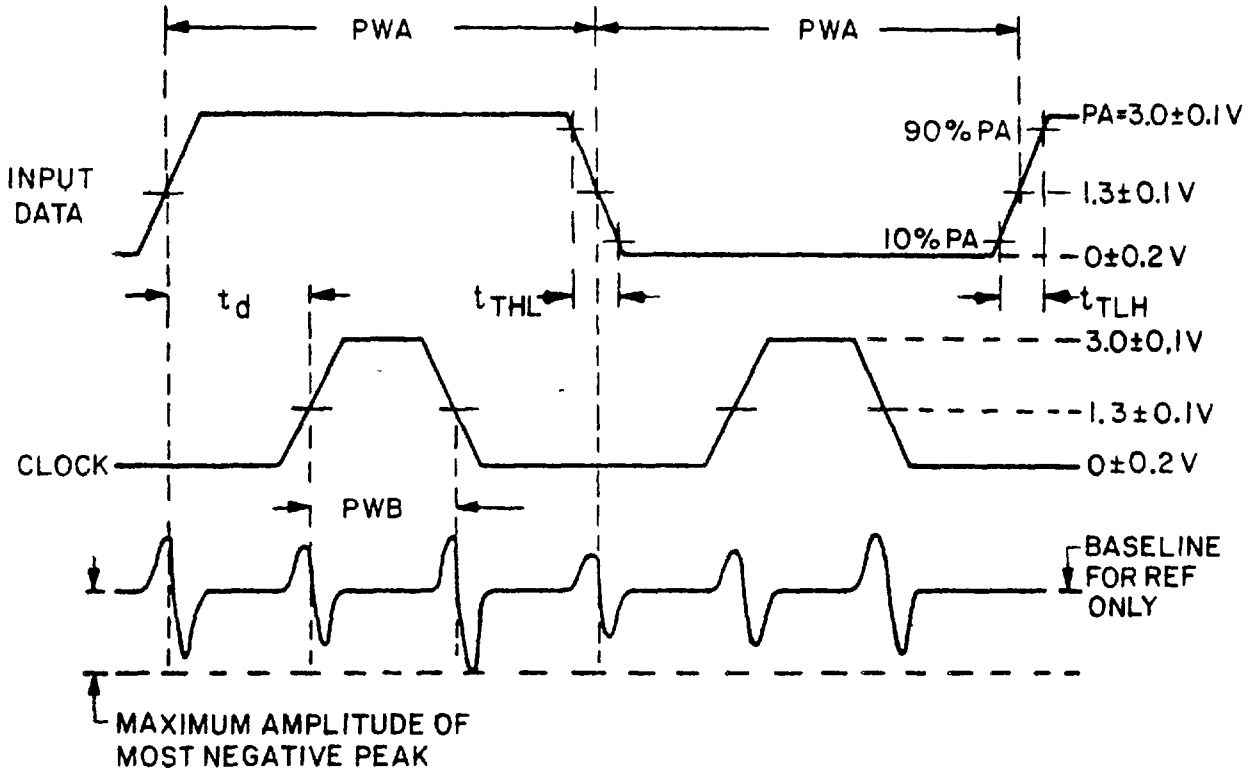
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NOTE. Input pulse parameters:
(a) Pulse width (PWA) = 500 ns ±10%.
(b) t_{TLH} = 3.5 ±1.5 ns (verified at each input pin).
(c) t_{THL} = 3.5 ±1.5 ns (verified at each input pin).
(d) Data repetition rate = 1.0 MHz ±10%.
(e) Clock pulse width (PWB) = 250 ns ±10%.
(f) Clock pulse repetition rate = 2.0 MHz ±10%.
(g) t_d = 200 ns ±10%.

FIGURE 16. Output crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

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NOTE: Input pulse parameters:
(a) Data pulse width (PWA) = 500 ns ±10%.
(b) t_{TLH} = 10 ±2 ns.
(c) t_{THL} = 10 ±2 ns.
(d) Data repetition rate = 1 MHz ±10%.
(e) Clock pulse width (PWB) = 250 ns ±10%.
(f) Clock repetition rate = 2 MHz ±10%.
(g) t_d = 200 ns ±10%.

FIGURE 17. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Propagation delay time (t_{PHL}) (set- and reset-to-output) test.

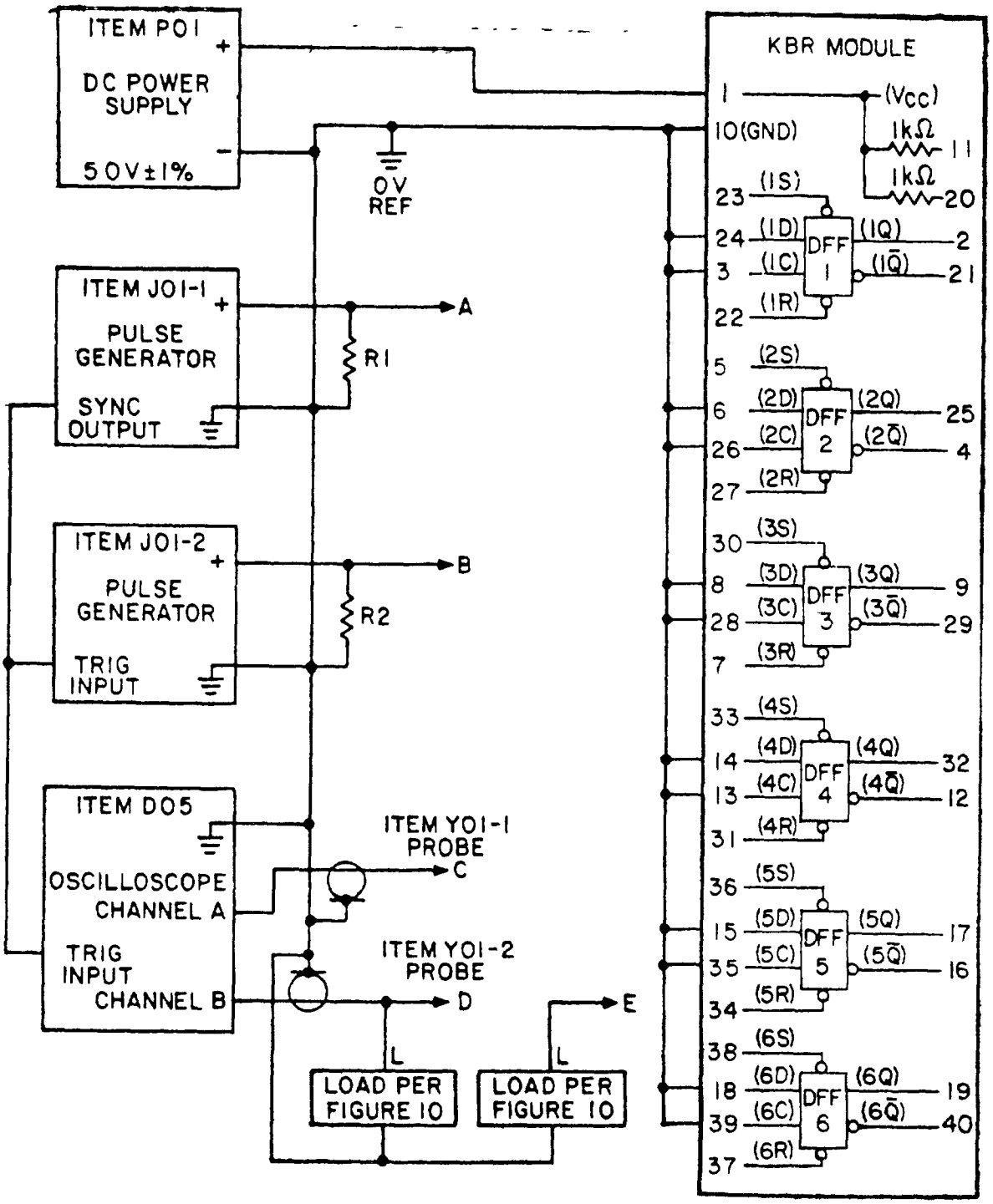
Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	1Q Propagation delay time (t_{PHL}), set and reset to output, C_L = 150 pF	23	22	B	2	21	10	200	ns	18,19
V2	$1\overline{Q}$	23	22	A	21	2				
V3	2Q	5	27	B	25	4				
V4	$2\overline{Q}$	5	27	A	4	25				
V5	3Q	30	7	B	9	29				
V6	$3\overline{Q}$	30	7	A	29	9				
V7	4Q	33	31	B	32	12				
V8	$4\overline{Q}$	33	31	A	12	32				
V9	5Q	36	34	B	17	16				
V10	$5\overline{Q}$	36	34	A	16	17				
V11	6Q	38	37	B	19	40				
V12	$6\overline{Q}$	38	37	A	40	19				
V13	1Q Propagation delay time (t_{PHL}), set and reset to output, C_L = 50 pF	23	22	B	2	21		150		
V14	$1\overline{Q}$	23	22	A	21	2				
V15	2Q	5	27	B	25	4				
V16	$2\overline{Q}$	5	27	A	4	25				
V17	3Q	30	7	B	9	29				
V18	$3\overline{Q}$	30	7	A	29	9				
V19	4Q	33	31	B	32	12				
V20	$4\overline{Q}$	33	31	A	12	32				
V21	5Q	36	34	B	17	16				
V22	$5\overline{Q}$	36	34	A	16	17				
V23	6Q	38	37	B	19	40				
V24	$6\overline{Q}$	38	37	A	40	19				

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TABLE XXI. Propagation delay time (t_{PLH})(set- and reset-to-output) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	1Q Propagation delay time (t_{PLH}), set and reset to output, C_L = 150 pF	23	22	A	2	21	10	125	ns	18,19
W2	$1\overline{Q}$	23	22	B	21	2				
W3	2Q	5	27	A	25	4				
W4	$2\overline{Q}$	5	27	B	4	25				
W5	3Q	30	7	A	9	29				
W6	$3\overline{Q}$	30	7	B	29	9				
W7	4Q	33	31	A	32	12				
W8	$4\overline{Q}$	33	31	B	12	32				
W9	5Q	36	34	A	17	16				
W10	$5\overline{Q}$	36	34	B	16	17				
W11	6Q	38	37	A	19	40				
W12	$6\overline{Q}$	38	37	B	40	19				
W13	1Q Propagation delay time (t_{PLH}), set and reset to output, C_L = 50 pF	23	22	A	2	21		75		
W14	$1\overline{Q}$	23	22	B	21	2				
W15	2Q	5	27	A	25	4				
W16	$2\overline{Q}$	5	27	B	4	25				
W17	3Q	30	7	A	9	29				
W18	$3\overline{Q}$	30	7	B	29	9				
W19	4Q	33	31	A	32	12				
W20	$4\overline{Q}$	33	31	B	12	32				
W21	5Q	36	34	A	17	16				
W22	$5\overline{Q}$	36	34	B	16	17				
W23	6Q	38	37	A	19	40				
W24	$6\overline{Q}$	38	37	B	40	19				

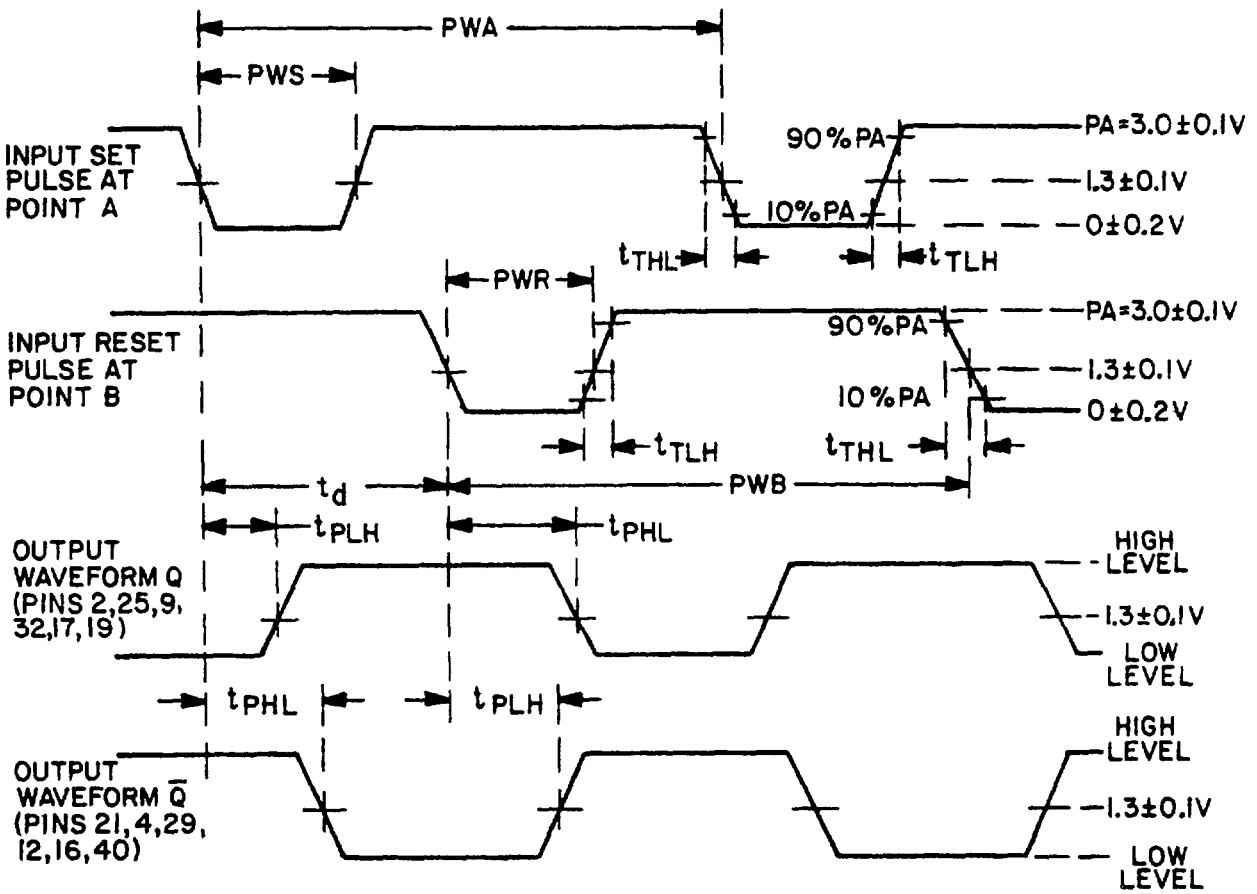
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NOTE: R1-R2 = 51Ω ±5%, 1 W.

FIGURE 18. Propagation delay time (t_{PHL} and t_{PLH}) (set- and reset-to-output) test circuit for tables XX and XXI.

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NOTES:
1. Adjust items J01-1 and J01-2 to provide input pulse parameters as follows
   (a) PWS = PWR = 100 $^{+2}_{-0}$ ns
   (b) PWA = PWB = 400 ±40 ns
   (c) t_d = 200 ±20 ns
   (d) t_{TLH} = 10 ±2 ns
   (e) t_{THL} = 10 ±2 ns
2. All voltage levels are referenced to ground.

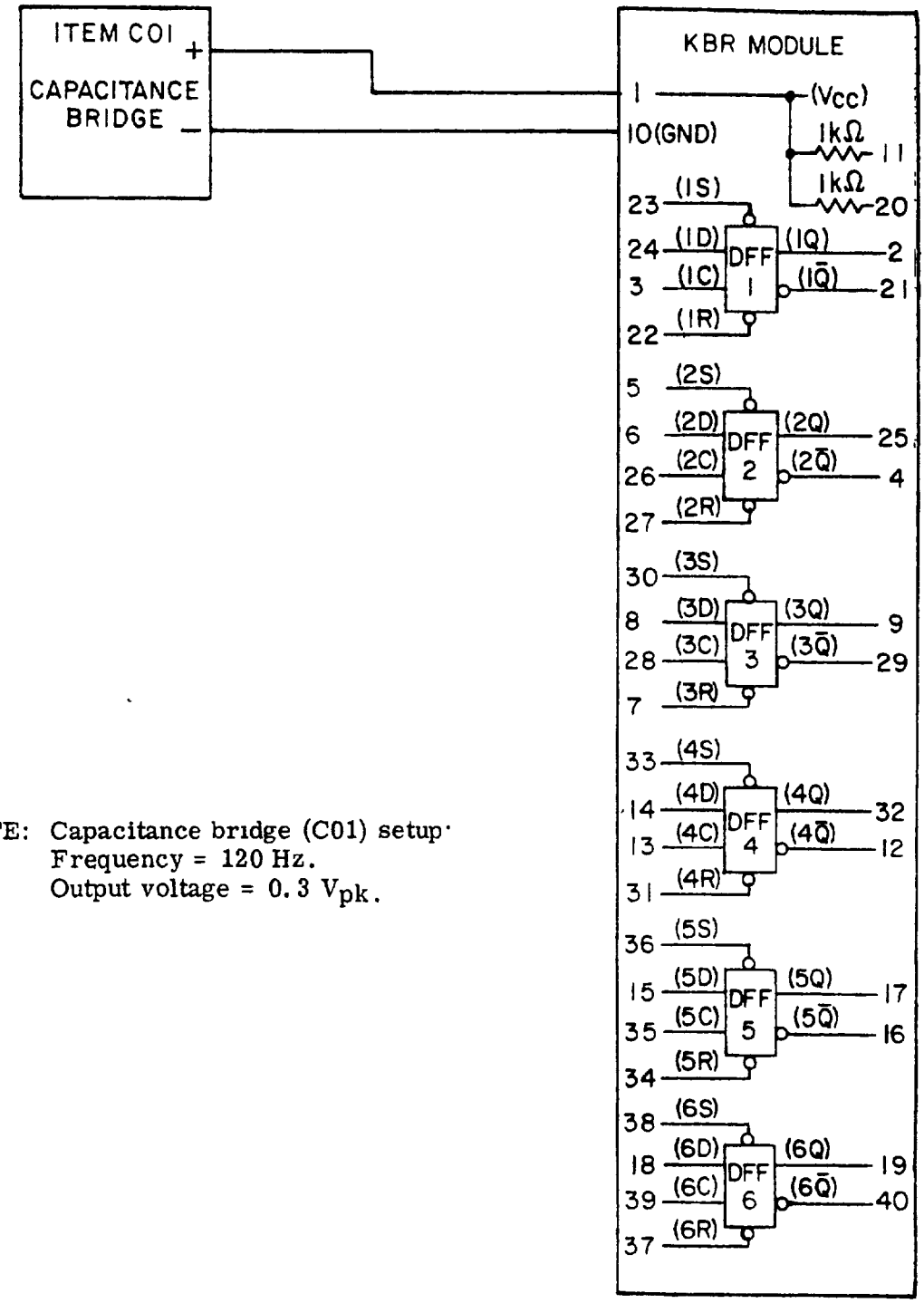
FIGURE 19. Propagation delay time (t_{PHL} and t_{PLH}) (set-and reset-to-output) test circuit waveforms for tables XX and XXI.

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TABLE XXII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
X	Power supply filter capacitance	1.6	2.8	µF	20

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NOTE: Capacitance bridge (C01) setup:
Frequency = 120 Hz.
Output voltage = 0.3 V_{pk}.

FIGURE 20. Power supply filter capacitance test circuit for table XXII.

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